Module 3: High-Dimensional Regression Models

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Introduction

In praise of linear models

- Recall the linear model

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \epsilon$$

- Despite its simplicity, the linear model has distinct advantages in terms of its interpretability and often shows good predictive performance.
- Hence we discuss in this lecture some ways in which the simple linear model can be improved, by replacing ordinary least squares fitting with some alternative fitting procedures.

Why consider alternatives to least squares?

- Prediction Accuracy: especially when $p > n$, to control the variance.
- Model Interpretability: By removing irrelevant features — that is, by setting the corresponding coefficient estimates to zero — we can obtain a model that is more easily interpreted. We will present some approaches for automatically performing feature selection.

Three classes of methods

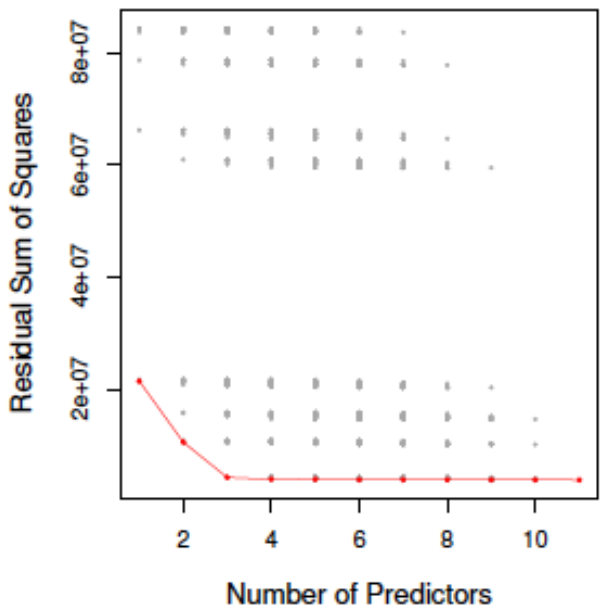
- Subset Selection. We identify a subset of the p predictors that we believe to be related to the response. We then fit a model using least squares on the reduced set of variables.
- Shrinkage. We fit a model involving all p predictors, but the estimated coefficients are shrunken towards zero relative to the least squares estimates. This shrinkage (also known as regularization) has the effect of reducing variance and can also perform variable selection.
- Dimension Reduction. We project the p predictors into a M-dimensional subspace, where $M < p$. This is achieved by computing M different linear combinations, or projections, of the variables. Then these M projections are used as predictors to fit a linear regression model by least squares.

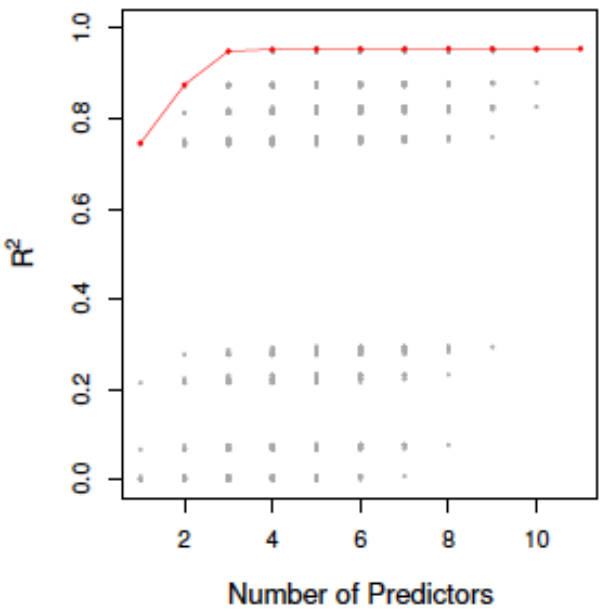
Subset Selection

Best subset selection

- Best subset and stepwise model selection procedures
- Best subset selection:
  1. Let $\mathcal{M}_0$ denote the null model, which contains no predictors. This model simply predicts the sample mean for each observation.
  2. For $k = 1, 2, \ldots, p$:
     —(a) For all $\binom{p}{k}$ models that contain exactly k predictors;
     —(b) Pick the best among these $\binom{p}{k}$ models, and call it $\mathcal{M}_k$. Here best is defined as having the smallest RSS, or equivalently largest R^2.
  3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using cross-validated prediction error, C_p, AIC, BIC, or adjusted R^2.

Example: credit data





For each possible model containing a subset of the ten predictors in the Credit data set, the RSS and R^2 are displayed. The red frontier tracks the best model for a given number of predictors, according to RSS and R^2. Though the data set contains only ten predictors, the x-axis ranges from 1 to 11, since one of the variables is categorical and takes on three values, leading to the creation of two dummy variables

Stepwise selection

- For computational reasons, best subset selection cannot be applied with very large p. Why not?
- Best subset selection may also suffer from statistical problems when p is large: the larger the search space, the higher the chance of finding models that look good on the training data, even though they might not have any predictive power on future data.
- Thus an enormous search space can lead to overfitting and high variance of the coefficient estimates.
- For both of these reasons, stepwise methods, which explore a far more restricted set of models, are attractive alternatives to best subset selection.

Forward stepwise selection

- Forward stepwise selection begins with a model containing no predictors, and then adds predictors to the model, one-at-a-time, until all of the predictors are in the model.
- In particular, at each step the variable that gives the greatest additional improvement to the fit is added to the model.

Forward stepwise selection

1. Let $\mathcal{M}_0$ denote the null model, which contains no predictors. This model simply predicts the sample mean for each observation.
2. For $k = 0, 1, 2, \ldots, p-1$:
   —(a) Consider all $p - k$ models that augment the predictors in $\mathcal{M}_k$ with one additional predictor;
   —(b) Pick the best among these $p - k$ models, and call it $\mathcal{M}_{k+1}$. Here best is defined as having the smallest RSS, or equivalently largest R^2.
3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using cross-validated prediction error, C_p, AIC, BIC, or adjusted R^2.

- Computational advantage over best subset selection is clear.
- It is not guaranteed to find the best possible model out of all 2^p models containing subsets of the p predictors. Why not?

Credit data example

# Variables	Best subset	Forward stepwise
One	rating	rating
Two	rating, income	rating, income
Three	rating, income, student	rating, income, student
Four	cards, income student, limit	rating, income, student, limit

The first four selected models for best subset selection and forward stepwise selection on the Credit data set. The first three models are identical but the fourth models differ.

Backward stepwise selection

- Like forward stepwise selection, backward stepwise selection provides an efficient alternative to best subset selection.
- However, unlike forward stepwise selection, it begins with the full least squares model containing all p predictors, and then iteratively removes the least useful predictor, one-at-a-time.

Forward stepwise selection

1. Let $\mathcal{M}_p$ denote the full model, which contains all p predictors.
2. For $k = p, p-1, \ldots, 1$:
   —(a) Consider all k models that contain all but one of the predictors in $\mathcal{M}_k$ for a total of $k-1$ predictor;
   —(b) Pick the best among these k models, and call it $\mathcal{M}_{k-1}$. Here best is defined as having the smallest RSS, or equivalently largest R^2.
3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using cross-validated prediction error, C_p, AIC, BIC, or adjusted R^2.

- Like forward stepwise selection, the backward selection approach searches through only $1 + p(p + 1)/2$ models, and so can be applied in settings where p is too large to apply best subset selection
- Like forward stepwise selection, backward stepwise selection is not guaranteed to yield the best model containing a subset of the p predictors.
- Backward selection requires that the number of samples n is larger than the number of variables p (so that the full model can be fit). In contrast, forward stepwise can be used even when $n < p$, and so is the only viable subset method when p is very large.

Choosing the optimal model

- The model containing all of the predictors will always have the smallest RSS and the largest R^2, since these quantities are related to the training error.
- We wish to choose a model with low test error, not a model with low training error. Recall that training error is usually a poor estimate of test error.
- Therefore, RSS and R^2 are not suitable for selecting the best model among a collection of models with different numbers of predictors.

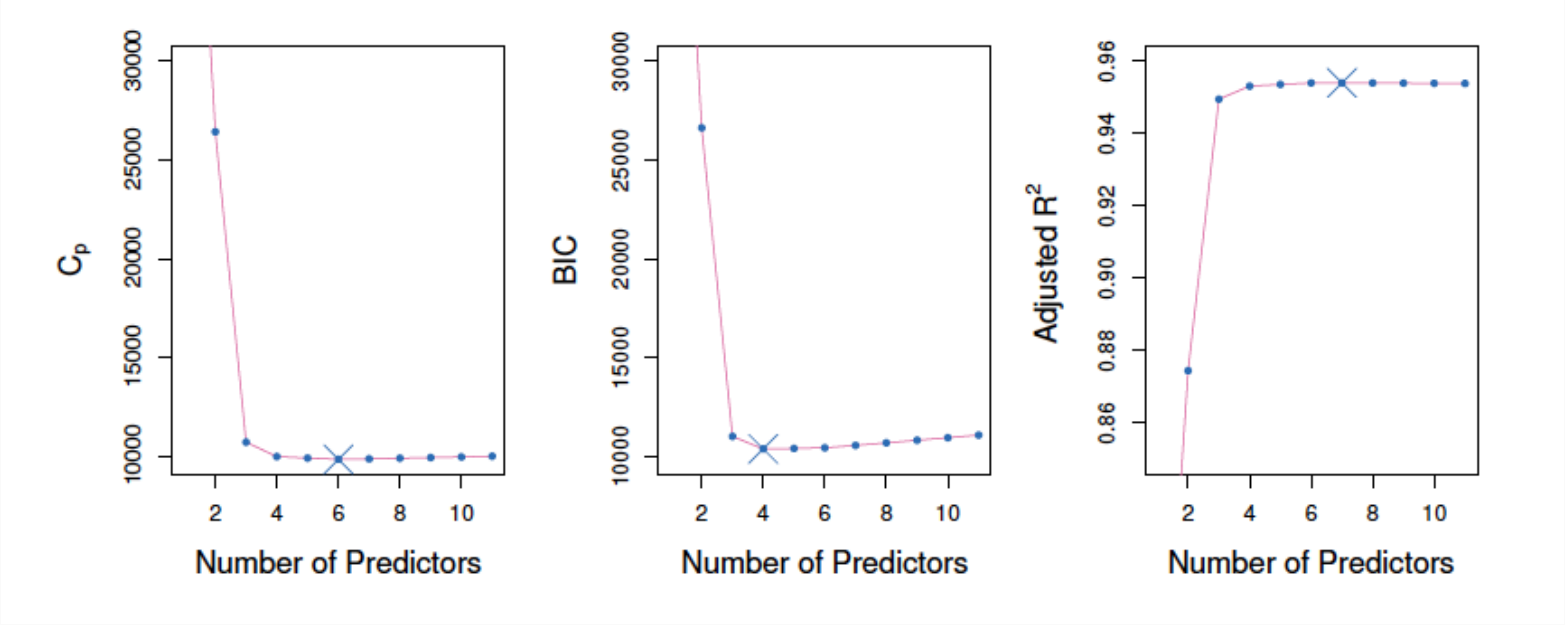
Estimating test error: two approaches

- We can indirectly estimate test error by making an adjustment to the training error to account for the bias due to overfitting.
- We can directly estimate the test error, using either a validation set approach or a cross-validation approach, as discussed in previous lectures.
- We illustrate both approaches next.

C_p, AIC, BIC, and adjusted R^2

- These techniques adjust the training error for the model size, and can be used to select among a set of models with different numbers of variables.
- The next figure displays C_p, BIC, and adjusted R^2 for the best model of each size produced by best subset selection on the Credit data set.

Credit data example



Some details

- Mallows's C_p:

$$C_p = \frac{1}{n}(RSS + 2d\hat{\sigma}^2),$$

  where d is the total number of parameters used and $\hat{\sigma}^2$ is an estimate of the variance of the error ϵ associated with each response measurement.

- The AIC criterion is defined for a large class of models fit by maximum likelihood:

$$AIC = -2 \log L + 2 \cdot d,$$

  where L is the maximized value of the likelihood function for the estimated model.

- In the case of linear model with Gaussian errors, maximum likelihood and least squares are the same thing, and C_p and AIC are equivalent. Prove this.

BIC

$$BIC = \frac{1}{n}(RSS + \log(n) \cdot d \cdot \hat{\sigma}^2)$$

- Like C_p, the BIC will tend to take on a small value for a model with a low test error, and so generally we select the model that has the lowest BIC value.
- Notice that BIC replaces the $2d\hat{\sigma}^2$ used by C_p with a $\log(n)d\hat{\sigma}^2$ term, where n is the number of observations.
- Since $\log n > 2$ for any $n > 7$, the BIC statistic generally places a heavier penalty on models with many variables, and hence results in the selection of smaller models than C_p. See the previous figure.

Adjusted R^2

- For a least squares model with d variables, the adjusted R^2 statistic is calculated as

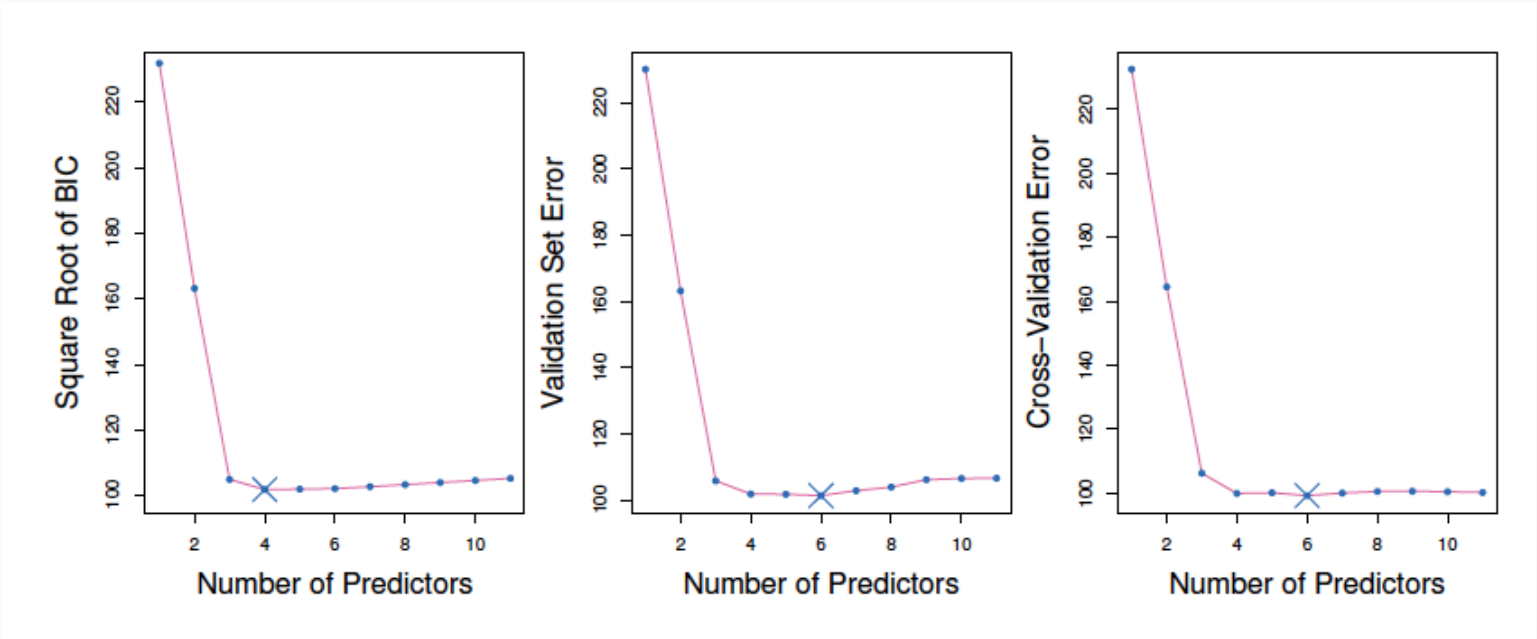
$$adj\ R^2 = 1 - \frac{RSS/(n-d-1)}{TSS/(n-1)},$$

  where TSS is the total sum of squares.
- Unlike C_p, AIC, and BIC, for which a small value indicates a model with a low test error, a large value of adjusted R^2 indicates a model with a small test error.
- Maximizing the adjusted R^2 is equivalent to minimizing $RSS/(n-d-1)$. While RSS always decreases as the number of variables in the model increases, $RSS/(n-d-1)$ may increase or decrease, due to the presence of d in the denominator.
- Unlike the R^2 statistic, the adjusted R^2 statistic pays a price for the inclusion of unnecessary variables in the model. See the previous figure.

Validation and cross-validation

- Each of the procedures returns a sequence of models $\mathcal{M}_k$ indexed by model size $k = 0, 1, 2; , \ldots$. Our job here is to select $\hat{k}$. Once selected, we will return model $\mathcal{M}_{\hat{k}}$.
- We compute the validation set error or the cross-validation error for each model $\mathcal{M}_k$ under consideration, and then select the k for which the resulting estimated test error is smallest.
- This procedure has an advantage relative to AIC, BIC, C_p, and adjusted R_2, in that it provides a direct estimate of the test error, and doesn't require an estimate of the error variance σ^2.
- It can also be used in a wider range of model selection tasks, even in cases where it is hard to pinpoint the model degrees of freedom (e.g. the number of predictors in the model) or hard to estimate the error variance σ^2.

Credit data example



- The validation errors were calculated by randomly selecting three-quarters of the observations as the training set, and the remainder as the validation set.
- The cross-validation errors were computed using $k = 10$ folds. In this case, the validation and cross-validation methods both result in a six-variable model.
- However, all three approaches suggest that the four-, five-, and six-variable models are roughly equivalent in terms of their test errors.
- In this setting, we can select a model using the one-standard-error rule. We first calculate the standard error of the estimated test MSE for each model size, and then select the smallest model for which the estimated test error is within one standard error of the lowest point on the curve. What is the rationale for this?

Shrinkage Method: Ridge

Shrinkage methods

Ridge regression and Lasso

- The subset selection methods use least squares to fit a linear model that contains a subset of the predictors.
- As an alternative, we can fit a model containing all p predictors using a technique that constrains or regularizes the coefficient estimates, or equivalently, that shrinks the coefficient estimates towards zero.
- It may not be immediately obvious why such a constraint should improve the fit, but it turns out that shrinking the coefficient estimates can significantly reduce their variance.

Variable selection

Given training set $(\mathbf{x}_i, y_i)_{i=1}^n$ with $y_i \in \mathbb{R}$ and $\mathbf{x}_i \in \mathbb{R}^p$, it is assumed that

$$Y = \beta_0 + \sum_{j=1}^{p_0} \beta_j x_j + \epsilon,$$

where $p_0 \ll p$, and sometimes $n \ll p$.

- $\mathcal{A}^* = \{1, \ldots, p_0\}$ indexes the informative covariates, and $\{p_0 + 1, \ldots, p\}$ indexes the redundant/uninformative covariates.
- The goal of variable selection is to correctly detect $\mathcal{A}^*$ from $\{1, \ldots, p\}$.
- Linear model is often assumed in large-p-small-n scenario, while detecting nonlinear relationship remains open.

Conventional/modern technique

- Subset selection
  - Various information criteria, cross validation, ...
  - Too many subset to compare.
- Sequential variable selection
  - Forward/backward selection
- Multiple testing
  - Focus on marginal correlation and may not work well in general
- Shrinkage method
  - Lasso and its variants
- Bayesian variable selection

Shrinkage methods

Shrinkage methods are more popular, and are formulated as

$$(\hat{\beta}_0, \hat{\beta}) = \underset{\beta}{\operatorname{argmin}} \sum_{i=1}^{n} (y_i - \beta_0 - x_i^T \beta)^2 + \lambda J(\beta).$$

Various choices of $J(\beta)$ lead to different shrinkage methods and possess different properties.

After centralization, it becomes

$$\hat{\beta} = \underset{\beta}{\operatorname{argmin}} \sum_{i=1}^{n} (y_i - x_i^T \beta)^2 + \lambda J(\beta).$$

Ridge regression

$$\hat{\beta}^{ridge} = \underset{\beta}{\operatorname{argmin}} \ (\mathbf{y} - \mathbf{X}\,\beta)^T(\mathbf{y} - \mathbf{X}\,\beta) + \lambda\|\beta\|^2$$

or equivalently,

$$\begin{aligned}\hat{\beta}^{ridge} &= \underset{\beta}{\operatorname{argmin}} \ (\mathbf{y} - \mathbf{X}\,\beta)^T(\mathbf{y} - \mathbf{X}\,\beta) \\ &\quad \text{subject to } \|\beta\|^2 \leq s.\end{aligned}$$

Solution is

$$\hat{\beta}_\lambda = (\mathbf{X}^T\mathbf{X} + \lambda I_p)^{-1}\mathbf{X}^T\mathbf{y}\,.$$

Note $\lambda = 0$ gives the least squares estimator, and if $\lambda \to \infty$ then $\hat{\beta}_\lambda \to 0$. In general, $\hat{\beta}_\lambda$ is a biased estimator that may have smaller mean squared error than the least squares estimator.

- As with least squares, ridge regression seeks coefficient estimates that fit the data well, by making the RSS small.
- However, the second term, $\lambda \sum_j \beta_j^2$, called a shrinkage penalty, is small when $\beta_1, \ldots, \beta_p$ are close to zero, and so it has the effect of shrinking the estimates of coefficients towards zero.
- The tuning parameter λ serves to control the relative impact of these two terms on the regression coefficient estimates.
- Selecting a good value for λ is critical; cross-validation is used for this.

Ridge regression

The ridge solution can be rewritten as

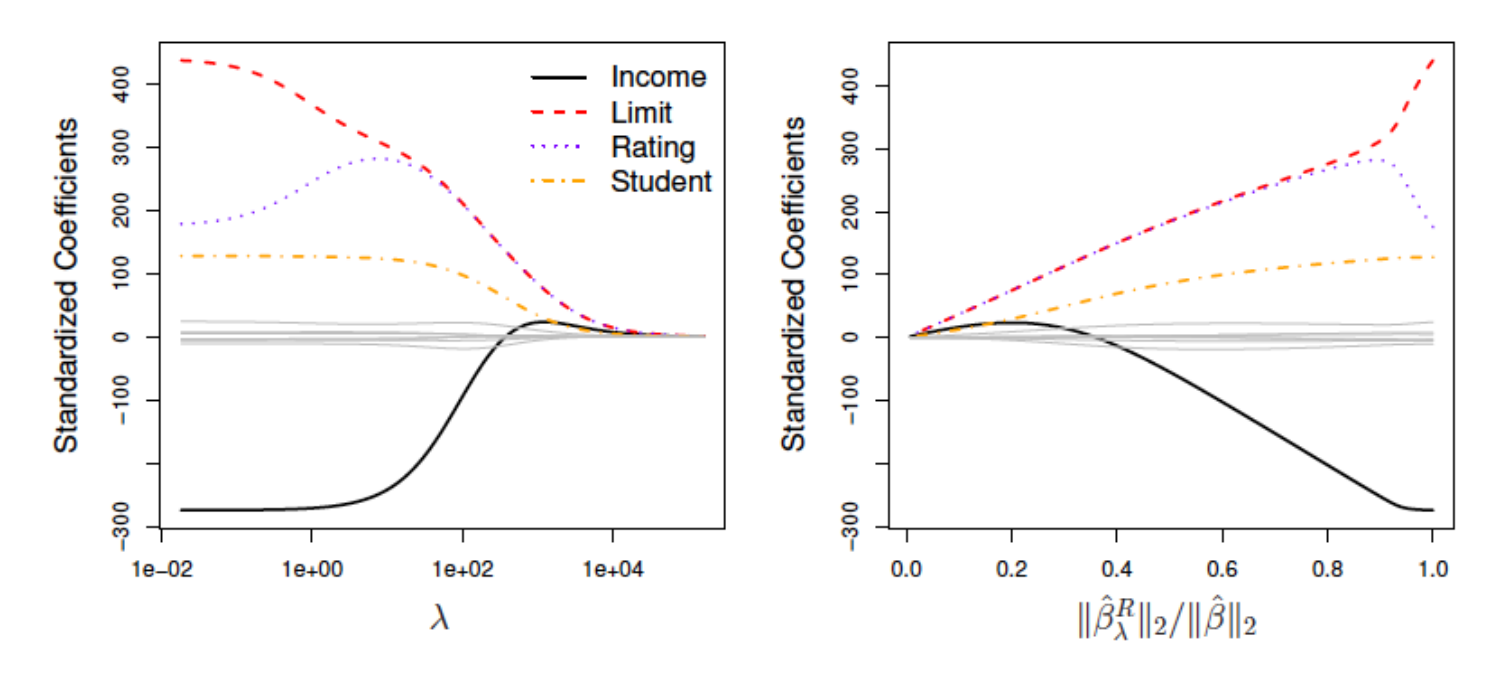
$$\begin{aligned}\hat{\mathbf{f}} &= \mathbf{X}\,\hat{\beta}^{ridge} = \mathbf{X}(\mathbf{X}^T\mathbf{X}+\lambda I_p)^{-1}\mathbf{X}^T\mathbf{y} \\ &= \mathbf{U}\,\mathbf{D}(\mathbf{D}^T\mathbf{D}+\lambda I_p)^{-1}\,\mathbf{D}\,\mathbf{U}^T\mathbf{y} \\ &= \sum_{j=1}^{p}\Big(\frac{d_j^2}{d_j^2+\lambda}\Big)\mathbf{u}_j\mathbf{u}_j^T\mathbf{y},\end{aligned}$$

where $\mathbf{X} = \mathbf{U}\,\mathbf{D}\,\mathbf{V}^T$ is a singular value decomposition of $\mathbf{X}$.

The effective degree of freedom of the ridge regression is

$$df(\lambda) = \mathrm{tr}\big(\mathbf{X}(\mathbf{X}^T\mathbf{X}+\lambda I_p)^{-1}\mathbf{X}^T\big) = \sum_{j=1}^{p}\frac{d_j^2}{d_j^2+\lambda}.$$

Credit data example



Details of previous figure

- In the left-hand panel, each curve corresponds to the ridge regression coefficient estimate for one of the ten variables, plotted as a function of λ.
- The right-hand panel displays the same ridge coefficient estimates as the left-hand panel, but instead of displaying λ on the x-axis, we now display $\|\hat{\beta}^R_\lambda\|_2/\|\hat{\beta}\|_2$, where $\hat{\beta}$ denotes the vector of least squares coefficient estimates.

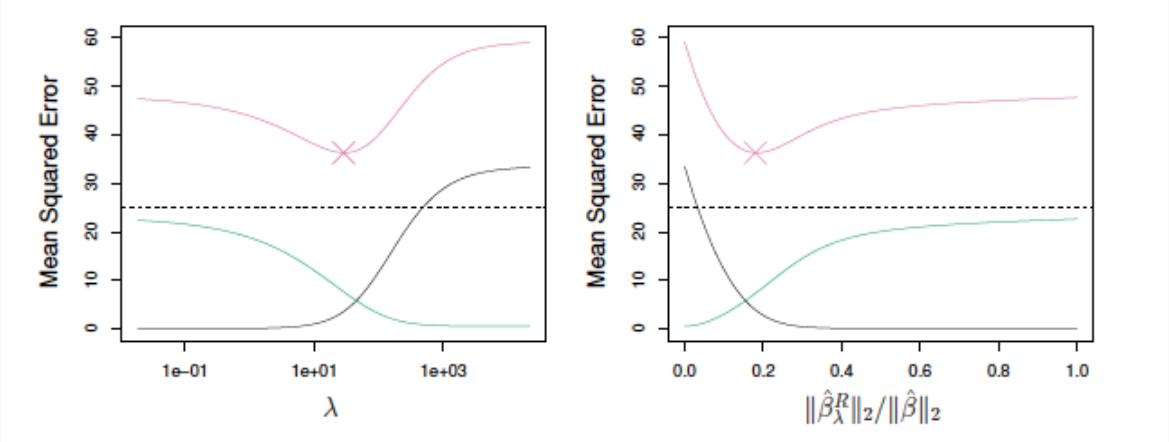
Scaling of predictors

- The standard least squares coefficient estimates are scale equivariant: multiplying X_j by a constant c simply leads to a scaling of the least squares coefficient estimates by a factor of $1/c$. In other words, regardless of how the jth predictor is scaled, $X_j\hat{\beta}_j$ will remain the same.
- In contrast, the ridge regression coefficient estimates can change substantially when multiplying a given predictor by a constant, due to the sum of squared coefficients term in the penalty part of the ridge regression objective function.
- Therefore, it is best to apply ridge regression after standardizing the predictors, using the formula

$$\tilde{x}_{ij} = \frac{x_{ij}}{\sqrt{\frac{1}{n}\sum_i (x_{ij} - \bar{x}_j)^2}}$$

Why ridge improves over LSE?

The bias-variance tradeoff



Simulated data with n = 50 observations, p = 45 predictors, all having nonzero coefficients. Squared bias (black), variance (green), and test mean squared error (purple) for the ridge regression predictions on a simulated data set, as a function of λ and $\|\hat{\beta}^R_\lambda\|_2/\|\hat{\beta}\|_2$. The horizontal dashed lines indicate the minimum possible MSE. The purple crosses indicate the ridge regression models for which the MSE is smallest.

Shrinkage Method: LASSO

The LASSO

- Ridge regression does have one obvious disadvantage: unlike subset selection, which will generally select models that involve just a subset of the variables, ridge regression will include all p predictors in the final model.
- The Lasso is an alternative to ridge regression that overcomes this disadvantage.
- In statistical parlance, the lasso uses an ℓ_1 penalty instead of an ℓ_2 penalty. The ℓ_1 norm of a coefficient vector β is given by $\|\beta\|_1 = \sum |\beta_j|$.

- As with ridge regression, the lasso shrinks the coefficient estimates towards zero.
- However, in the case of the lasso, the ℓ_1 penalty has the effect of forcing some of the coefficient estimates to be exactly equal to zero when the tuning parameter λ is sufficiently large.
- Hence, much like best subset selection, the lasso performs variable selection.
- We say that the lasso yields sparse models — that is, models that involve only a subset of the variables.
- As in ridge regression, selecting a good value of λ for the lasso is critical; cross-validation is again the method of choice.

Lasso

$$\hat{\beta}^{lasso} = \underset{\beta}{\operatorname{argmin}} \ (\mathbf{y} - \mathbf{X}\,\beta)^T(\mathbf{y} - \mathbf{X}\,\beta) + \lambda\|\beta\|_1$$

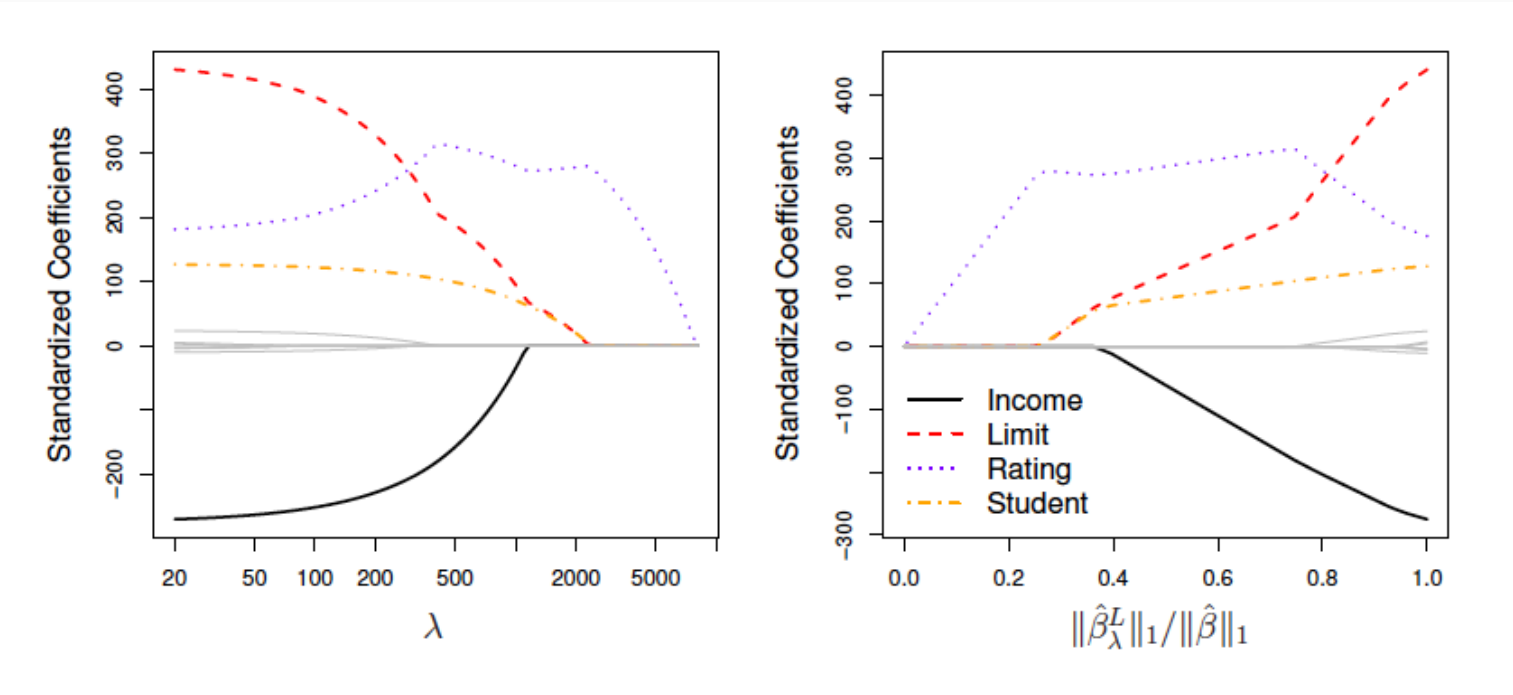
or equivalently,

$$\begin{aligned}\hat{\beta}^{lasso} \quad = \quad & \underset{\beta}{\operatorname{argmin}} \ (\mathbf{y} - \mathbf{X}\,\beta)^T(\mathbf{y} - \mathbf{X}\,\beta) \\ & \text{subject to } \|\beta\|_1 \leq s,\end{aligned}$$

where $\|\beta\|_1 = \sum_{j=1}^{p} |\beta_j|$.

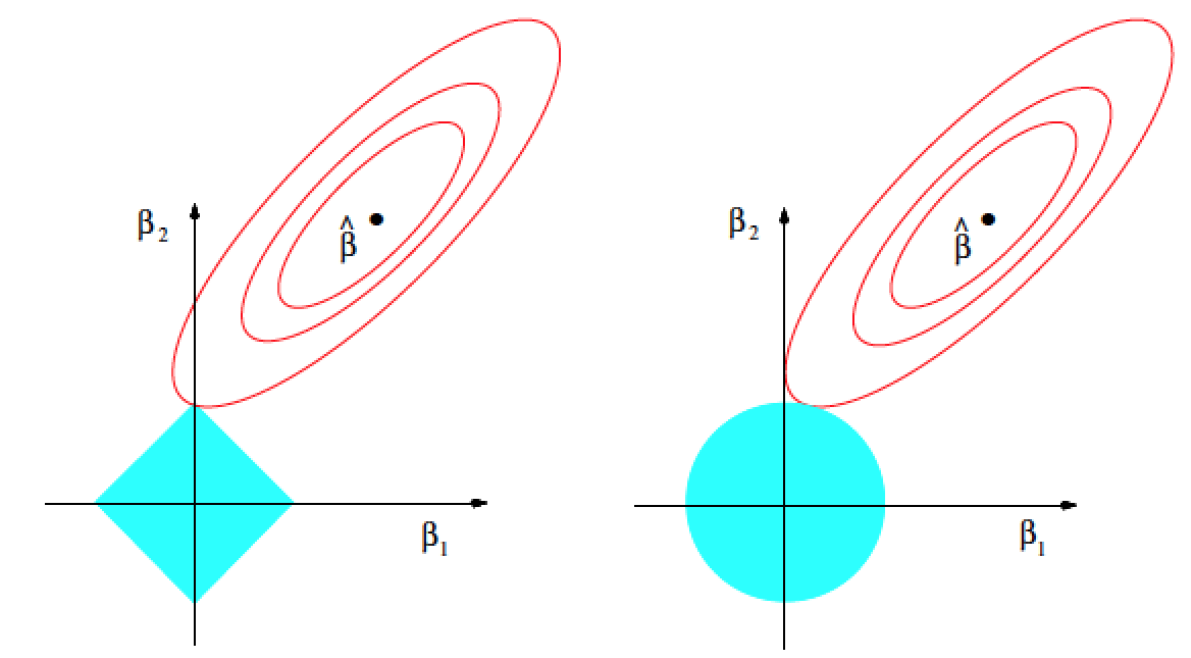
Notice that ridge penalty $\|\beta\|^2$ is replaced by $\|\beta\|_1$. This makes the solutions nonlinear in $\mathbf{y}$, and a quadratic programming algorithm cannot be used to solve the optimization problem.

Credit data example



Sparse solution

Due to the nature of the l_1-norm constraint, if s is small enough some coefficients of the lasso solution become exactly zero. Thus the lasso does some kind of continuous variable selection.



Solution path

- Clearly, the lasso solution $\hat{\beta}^{lasso}$ changes with the value of λ
- Even better, $\hat{\beta}^{lasso}$ is a piecewise linear function of λ
- The path algorithm starts at $\lambda = \infty$ or $s = 0$, and traces the solution path by successively solving the KKT conditions
- The key is to find the turning knots $\lambda_1, \ldots, \lambda_T$
- An interesting reading: Efron et al. (AOS; 2003)

- Path algorithms are available for many methods, such as fused lasso, trend filtering, locally adaptive regression splines, SVMs, 1-norm SVMs, relaxed maximum entropy method ...

Semi-differentiable functions

- One obvious challenge that comes with the lasso is that, by introducing absolute values, we are no longer dealing with differentiable functions.
- For this reason, we're going to take a moment and extend some basic calculus results to the case of non-differentiable (more specifically, semi-differentiable) functions.
- A function $f : \mathbb{R} \to \mathbb{R}$ is said to be semi-differentiable at a point x if both $d_- f(x)$ and $d_+ f(x)$ exist as real numbers, where $d_- f(x)$ and $d_+ f(x)$ are the left- and right-derivatives of f at x.
- Note that f is semi-differentiable implies that f is continuous.

Subderivatives and subdifferentials

- Give a semi-differentiable function $f : \mathbb{R} \to \mathbb{R}$, we say that d is a subderivative of f at x if $d \in [d_- f(x), d_+ f(x)]$; the set $[d_- f(x), d_+ f(x)]$ is called the subdifferential of f at x, and denoted by ∂f.
- Note that the subdifferential is a set-valued function.
- Recall that a function is differentiable at x if $d_- f(x) = d_+ f(x)$, i.e., if the subdifferential consists of a single point.
- Example: $f(x) = |x|$

Optimization

- The essential results of optimization can be extended to semi-differentiable functions.
- Theorem: If f is a semi-differentiable function and x_0 is a local minimum or maximum of f, then $0 \in \partial f(x_0)$.
- As with regular calculus, the converse is not true in general.

KKT conditions

- In the optimization literature, the resulting equations are known as the Karush-Kuhn-Tucker (KKT) conditions.
- For convex optimization problems such as the lasso, the KKT conditions are both necessary and sufficient to characterize the solution.
- A rigorous proof of this claim in multiple dimensions would involve some of the details we glossed over, but the idea is fairly straightforward: to solve for $\hat{\beta}$, we simply replace the derivative with the subderivative and the likelihood with the penalized likelihood.

KKT conditions for the lasso

- Result: $\hat{\beta}$ minimizes the lasso objective function if and only if it satisfies the KKT conditions

$$\frac{1}{n} x_j^T (y - X\hat{\beta}) = \lambda \, \text{sign}(\hat{\beta}_j), \qquad \hat{\beta}_j \neq 0$$

$$\frac{1}{n} |x_j^T (y - X\hat{\beta})| \leq \lambda, \qquad \hat{\beta}_j = 0$$

- In other words, the correlation between a predictor and the residuals, $x_j^T (y - X\hat{\beta})/n$, must exceed a certain minimum threshold λ before it is included in the model.
- When this correlation is below λ, $\hat{\beta}_j = 0$.

Remarks

- If we set
$$\lambda = \lambda_{max} = \max_{1 \leq j \leq p} |x_j^T y|/n,$$
then $\hat{\beta} = 0$ satisfies the KKT conditions.
- That is, for any $\lambda \geq \lambda_{max}$, we have $\hat{\beta}(\lambda) = 0$.
- On the other hand, if we set $\lambda = 0$, the KKT conditions are simple the normal equations for OLS, $X^T(y - X\hat{\beta}) = 0$.
- Thus, the coefficient path for the lasso starts at λ_{max} and may continue until $\lambda = 0$ if X is full rank; otherwise it will terminate at some $\lambda_{min} > 0$ when the model becomes saturated.

Lasso and uniqueness

- The lasso criterion is convex, but not strictly convex if $X^T X$ is not full rank; thus the lasso solution may not be unique.
- For example: Suppose $n = 2$ and $p = 2$ with $(y_1, x_{11}, x_{12}) = (1, 1, 1)$ and $(y_2, x_{21}, x_{22}) = (-1, -1, -1)$.
- Then the solutions are:

$$(\hat{\beta}_1, \hat{\beta}_2) = (0, 0), \quad \text{if } \lambda \geq 1$$

$$(\hat{\beta}_1, \hat{\beta}_2) \in \Big\{(\beta_1, \beta_2) : \beta_1 + \beta_2 = 1 - \lambda, \beta_1 \geq 0, \beta_2 \geq 0\Big\}, \quad \text{if } 0 \leq \lambda < 1$$

Lasso algorithms

- We introduced the lasso and derived necessary and sufficient conditions $\hat{\beta}$ must satisfy in order to minimize the lasso objective function.
- However, these conditions only allow us to check a solution; they do not necessarily help us to find the solution in the first place.
- We will discuss two algorithms for solving for $\hat{\beta}$; the algorithms are, of course, a practical necessity but also yield considerable insight into the nature of the lasso as a statistical method.

Forward selection and the lasso

- To get around the difficulty of finding the best possible subset, a common approach is to employ the greedy algorithm known as forward selection.
- Like forward selection, the lasso will allow more variables to enter the model as λ is lowered.
- However, the lasso performs a continuous version of variable selection and is less greedy about allowing selected variables into the model.

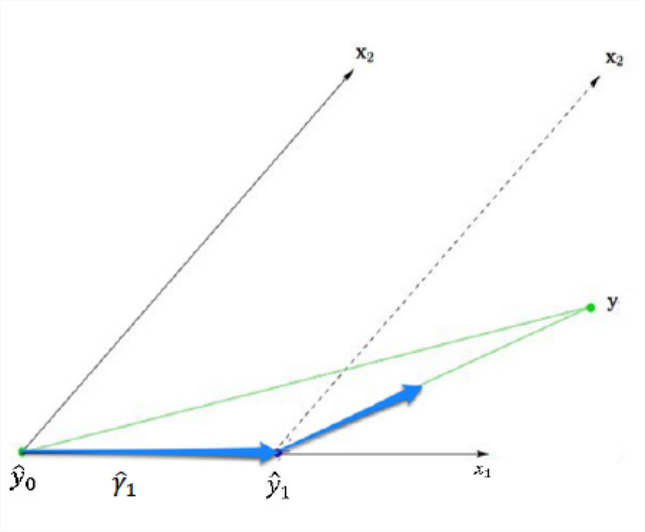
Forward selection and lasso paths

- Let us consider the regression paths of the lasso and forward selection as we lower λ, starting at λ_{max} where $\hat{\beta} = 0$.
- As λ is lowered below λ_{max}, lasso finds the predictor most highly correlated with the response (let x_j denote this predictor), and set $\hat{\beta}_j \neq 0$:
  — With forward selection, the estimate jumps from $\hat{\beta}_j = 0$ all the way to $\hat{\beta}_j = x_j^T y/n$
  — The lasso solution $\hat{\beta}_j = 0$ heads in this direction as well, but proceeds more cautiously, gradually advancing towards $\hat{\beta}_j = x_j^T y/n$ as we lower λ.

Remarks

- The lasso solution proceeds in this manner until it reaches the point that a new predictor, x_k, is equally correlated with the residual $r(\lambda) = y - X\hat{\beta}(\lambda)$.
- From this point, the lasso solution will contain both x_1 and x_2, and proceed in the direction that is equiangular between the two predictors.
- The lasso always proceeds in a direction such that every active predictor (i.e., one with $\hat{\beta}_j \neq 0$) is equally correlated with the residual $r(\lambda)$, which can also been seen from the KKT conditions.
- The geometry of the lasso clearly illustrates the "greediness" of forward selection.
- The lasso, meanwhile, allows the predictors most highly correlated with the residuals into the model, but only gradually, up to the point that the next predictor is equally useful in explaining the outcome.

Example



- Example with 2 correlated variables x_1 and x_2
- y is the response, $\hat{y}_0$,$\hat{y}_1$,$\hat{y}_2$ are estimated responses at each step, $\hat{\gamma}_1, \hat{\gamma}_2$ are the optimal step sizes
- At $\hat{y}_0 = 0$, the residual vector $y - \hat{y}_0$ is most correlated to x_1 (least angle)

$$\hat{y}_1 = \hat{y}_0 + \hat{\gamma}_1 x_1$$

- Select $\hat{\gamma}_1$ such that the residual $y - \hat{y}_1$ is equally correlated with x_1 and x_2
- Then, $\hat{y}_2 = \hat{y}_1 + \hat{\gamma}_2 u_2$, where u_2 is the unit bisector (equiangular vector)
- Here, $\hat{y}_2 = y$

LARS

- These geometric insights were the key to developing the first efficient algorithm for finding the lasso estimates $\hat{\beta}(\lambda)$.
- The approach, known as least angle regression, or the LARS algorithm, offers an elegant way to carry out lasso estimation.
- The idea behind the algorithm is to
  — (1) Project the residuals onto the active variables
  — (2) Calculate how far we can proceed in that direction before another variable reaches the necessary level of correlation with the residuals
  then adding it to the set of active variables and repeating (1) and (2), and so on.

Some details

- At kth step, let A_k be the set of active predictors and β_{A_k} be the coefficient vectors and $\hat{y}_k$ be the estimated response
- X_{A_k} be the active variables, i.e., those variables whose absolute correlation with the residual vector equals the maximal achievable absolute correlation
- The current residual will be $r_k = y - \hat{y}_k = y - X_{A_k}\beta_{A_k}$
- The coefficients are moved in the direction $\delta_k = (X_{A_k}^T X_{A_k})^{-1} X_{A_k}^T r_k$, i.e., $\beta_{A_k}(\gamma) = \beta_{A_k} + \gamma\delta_k$

- Thus $\hat{y}(\gamma) = X_{A_k}\beta_{A_k}(\gamma) = X_{A_k}\beta_{A_k} + \gamma X_{A_k}\delta_k$
- Prove that $u_k = X_{A_k}\delta_k$ makes equal angles with the active predictors

$$X_{A_k}^T u_k = X_{A_k}^T X_{A_k}\delta_k = X_{A_k}^T (X_{A_k}^T X_{A_k})^{-1} X_{A_k}^T r_k = X_{A_k}^T r_k$$

- Here, $u_k = r_k$, and r_k has equal correlation with all active predictors at kth step
- Thus, the elements of the vector $X_{A_k}^T r_k$ are all the same since the variables have equal correlation and make equal angles with the residual

LARS Algorithm

Algorithm 3.2 *Least Angle Regression.*

1. Standardize the predictors to have mean zero and unit norm. Start with the residual $\mathbf{r} = \mathbf{y} - \bar{\mathbf{y}}$, $\beta_1, \beta_2, \ldots, \beta_p = 0$.

2. Find the predictor $\mathbf{x}_j$ most correlated with $\mathbf{r}$.

3. Move β_j from 0 towards its least-squares coefficient $\langle \mathbf{x}_j, \mathbf{r} \rangle$, until some other competitor $\mathbf{x}_k$ has as much correlation with the current residual as does $\mathbf{x}_j$.

4. Move β_j and β_k in the direction defined by their joint least squares coefficient of the current residual on $(\mathbf{x}_j, \mathbf{x}_k)$, until some other competitor $\mathbf{x}_l$ has as much correlation with the current residual.

5. Continue in this way until all p predictors have been entered. After $\min(N - 1, p)$ steps, we arrive at the full least-squares solution.

Historical role of LARS

- The LARS algorithm played an important role in the history of the lasso.
- Prior to LARS, lasso estimation was slow and very computer intensive; LARS, on the other hand, requires only $O(np^2)$ calculations, the same order of magnitude as OLS.
- Nevertheless, LARS is not widely used anymore
- Instead, the most popular approach for fitting lasso and other penalized regression models is to employ coordinate descent algorithms, a less beautiful but simpler and more flexible alternative.

Coordinate descent

- The idea behind coordinate descent is, simply, to optimize a target function with respect to a single parameter at a time, iteratively cycling through all parameters until convergence is reached.
- Coordinate descent is particularly suitable for problems, like the lasso, that have a simple closed form solution in a single dimension but lack one in higher dimensions.

CD notation

- Let us consider minimizing Q with respect to β_j, while temporarily treating the other regression coeffcients β_{-j} as fixed:

$$Q(\beta_j|\beta_{-j}) = \frac{1}{2n}\sum_{i=1}^{n}(y_i - \sum_{k\neq j} x_{ij}\beta_k - x_{ij}\beta_j)^2 + \lambda|\beta_j| + \textit{constant}$$

- Let

$$\tilde{r}_{ij} = y_i - \sum_{k\neq j} x_{ik}\tilde{\beta}_k$$

$$\tilde{z}_j = n^{-1}\sum_{i=1}^{n} x_{ij}\tilde{r}_{ij},$$

where $\{\tilde{r}_{ij}\}_{i=1}^n$ are the partial residuals with respect to the jth predictor, and $\tilde{z}_j$ is the OLS estimator based on $\{\tilde{r}_{ij}, x_{ij}\}_{i=1}^n$

CD algorithm

- It is easy to solve the problem of finding a one-dimensional lasso solution; letting $\tilde{\beta}_j$ denote the minimizer of $Q(\beta_j|\tilde{\beta}_{-j})$,

$$\tilde{\beta}_j = S(\tilde{z}_j|\lambda)$$

- This suggest the following algorithm:

repeat

  for $j = 1, 2, \ldots, p$

    $\tilde{z}_j = n^{-1} \sum_{i=1}^{n} x_{ij} r_i + \widetilde{\beta}_j^{(s)}$

    $\widetilde{\beta}_j^{(s+1)} \leftarrow S(\tilde{z}_j|\lambda)$

    $r_i \leftarrow r_i - (\widetilde{\beta}_j^{(s+1)} - \widetilde{\beta}_j^{(s)}) x_{ij}$ for all i.

until convergence

Remarks

- The coordinate descent algorithm has the potential to be quite efficient, in that its three require only $O(2n)$ operations (no complicated matrix factorizations, or even matrix multiplication, just two inner products)
- Thus, one full iteration can be completed at a computational cost of $O(2np)$ operations
- Thus, coordinate descent is linear in both n and p, scaling up to high dimensions even better than LARS, although it is worth noting that coordinate descent requires an unknown number of iterations, whereas LARS terminates in a known number of steps

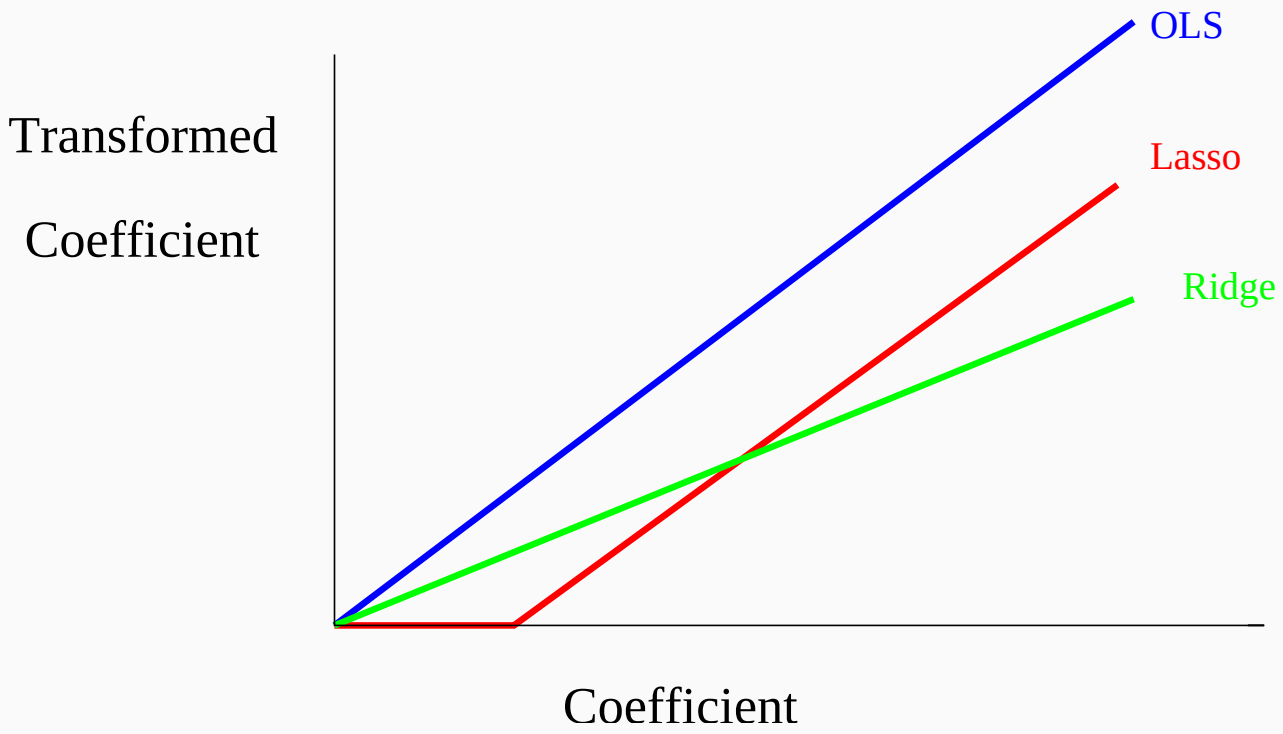
CD and pathwise optimization

- As we saw with ridge regression, we are typically interested in determining $\hat{\beta}$ for a range of values of λ, thereby obtaining the coefficient path
- In applying the coordinate descent algorithm to determine the lasso path, an efficient strategy is to compute solutions for decreasing values of λ, starting at λ_{max}, the point at which all coefficients are 0
- By continuing along a decreasing grid of λ values, we can use the solutions $\hat{\beta}(\lambda_k)$ as initial values when solving for $\hat{\beta}(\lambda_{k+1})$

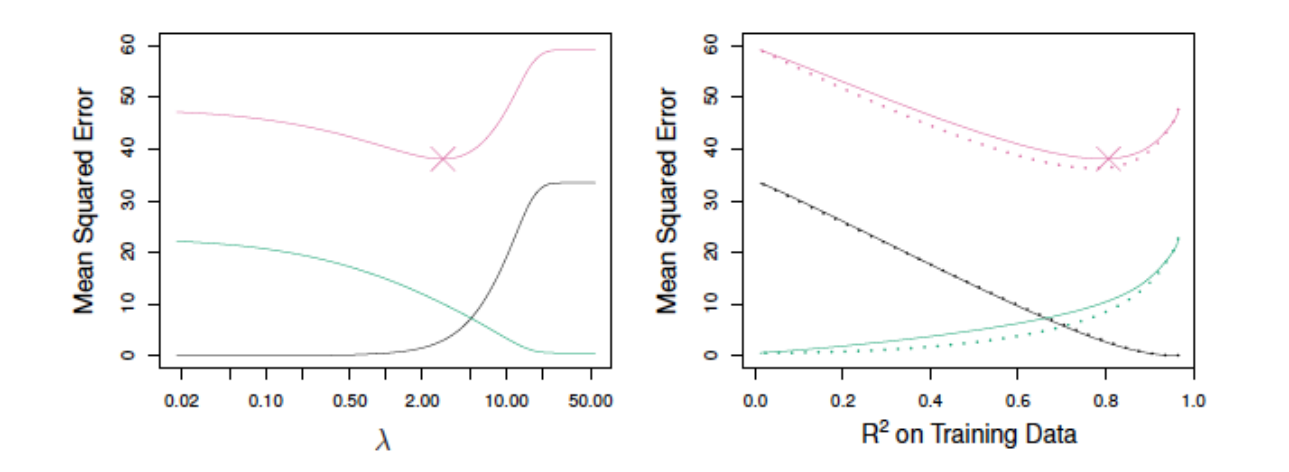
Ridge vs Lasso

If the columns of $\mathbf{X}$ are orthogonal, ridge multiplies least squares coefficients by a constant < 1, lasso truncates them towards zero by a constant.

- $\hat{\beta}_j^{ridge} = \hat{\beta}_j/(1+\lambda)$
- $\hat{\beta}_j^{lasso} = \text{sign}(\hat{\beta}_j)(|\hat{\beta}_j| - \lambda)_+$

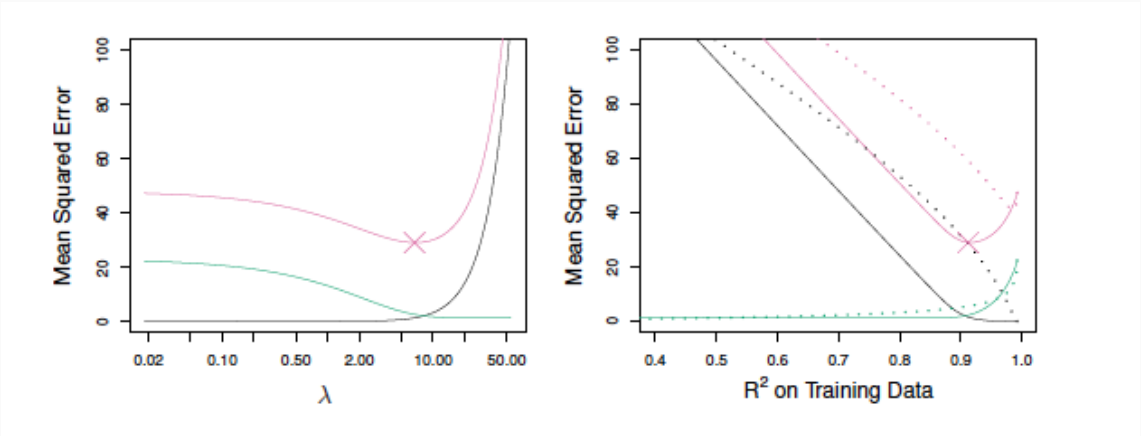


Ridge vs. Lasso



Left: Plots of squared bias (black), variance (green), and test MSE (purple) for the lasso on a previous simulated data set. Right: Comparison of squared bias, variance and test MSE between lasso (solid) and ridge (dashed). Both are plotted against their R^2 on the training data, as a common form of indexing. The crosses in both plots indicate the lasso model for which the MSE is smallest.

Ridge vs. Lasso



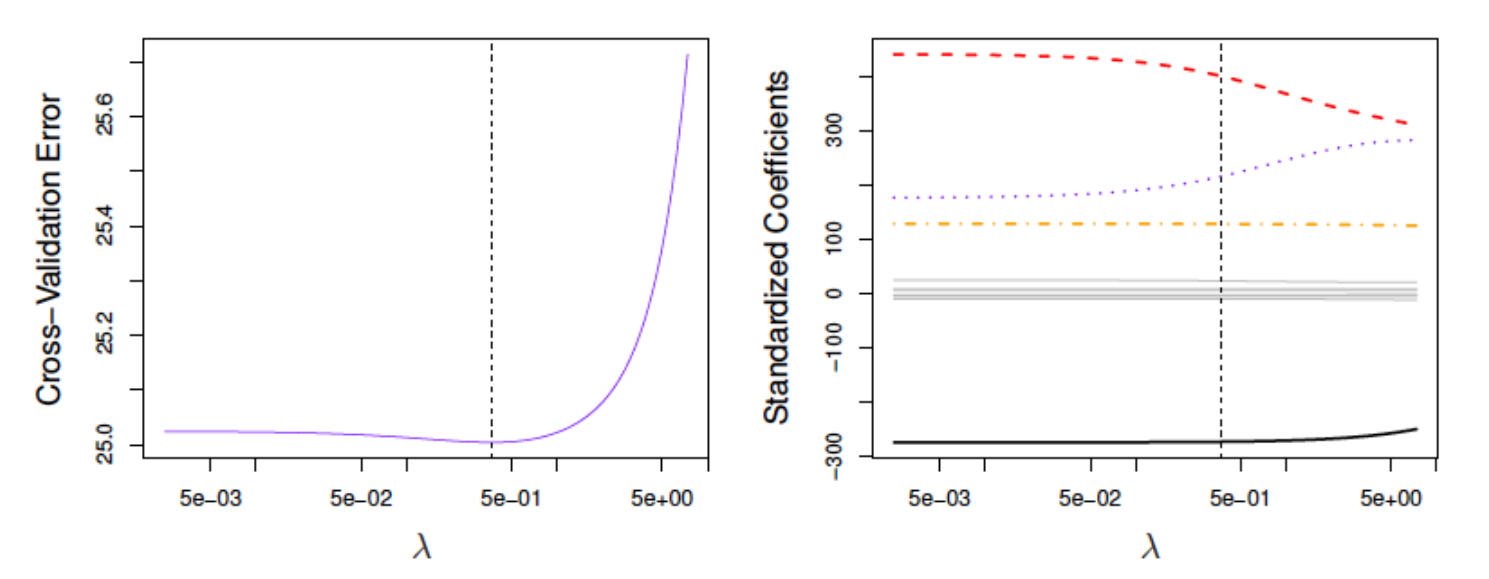
Left: Plots of squared bias (black), variance (green), and test MSE (purple) for the lasso. The simulated data is similar to the previous simulated data set, except that now only two predictors are related to the response. Right: Comparison of squared bias, variance and test MSE between lasso (solid) and ridge (dashed). Both are plotted against their R^2 on the training data, as a common form of indexing.

- These two examples illustrate that neither ridge regression nor the lasso will universally dominate the other.
- In general, one might expect the lasso to perform better when the response is a function of only a relatively small number of predictors.
- However, the number of predictors that is related to the response is never known a priori for real data sets.
- A technique such as cross-validation can be used in order to determine which approach is better on a particular data set.

Selecting the tuning parameter

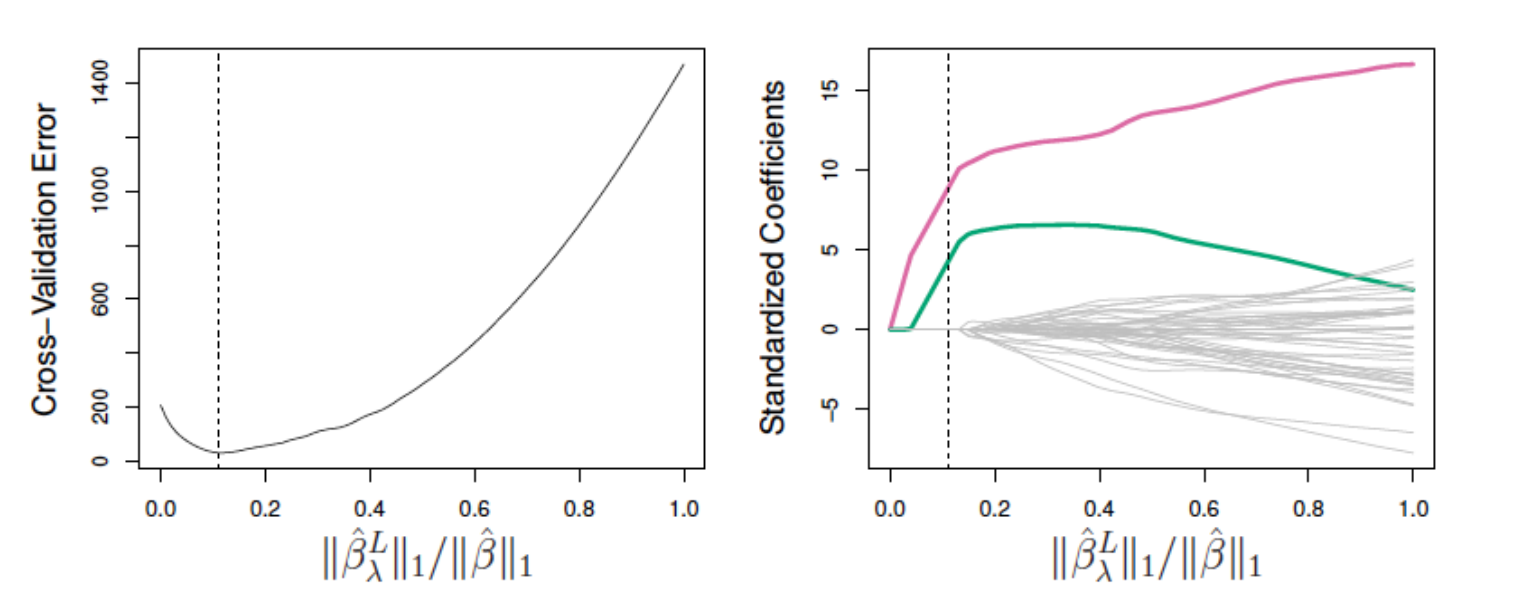
- As for subset selection, for ridge regression and lasso we require a method to determine which of the models under consideration is best.
- That is, we require a method selecting a value for the tuning parameter λ or equivalently, the value of the constraint s.
- Cross-validation provides a simple way to tackle this problem. We choose a grid of λ values, and compute the cross-validation error rate for each value of λ.
- We then select the tuning parameter value for which the cross-validation error is smallest.
- Finally, the model is re-fit using all of the available observations and the selected value of the tuning parameter.

Credit data example



Left: Cross-validation errors that result from applying ridge regression to the Credit data set with various values of λ. Right: The coefficient estimates as a function of λ. The vertical dashed lines indicates the value of λ selected by cross-validation.

Simulated data example



Left: Ten-fold cross-validation MSE for the lasso, applied to the sparse simulated data set. Right: The corresponding lasso coefficient estimates are displayed. The vertical dashed lines indicate the lasso fit for which the cross-validation error is smallest.

Other shrinkage methods

For $r \geq 0$, consider

$$\hat{\beta}^{bridge} = \underset{\beta}{\operatorname{argmin}} \ (\mathbf{y} - \mathbf{X}\beta)^T(\mathbf{y} - \mathbf{X}\beta)$$

$$\text{subject to } \sum_{j=1}^{p} |\beta_j|^r \leq s.$$

Bridge estimators

With $L_r(\beta) = \sum_{j=1}^{p} |\beta_j|^r$,

$$\hat{\beta}^{bridge} = \underset{\beta}{\operatorname{argmin}} \| \mathbf{y} - \mathbf{X}\,\beta \|^2 + \lambda L_r(\beta)$$

- $L_0(\beta) = \sum_{j=1}^{p} I(\beta_j \neq 0)$; (Hard thresholding)
- $L_1(\beta) = \sum_{j=1}^{p} |\beta_j|$; (Lasso)
- $L_2(\beta) = \sum_{j=1}^{p} \beta_j^2$; (Ridge regression)
- $L_\infty(\beta) = \max_j |\beta_j|$.

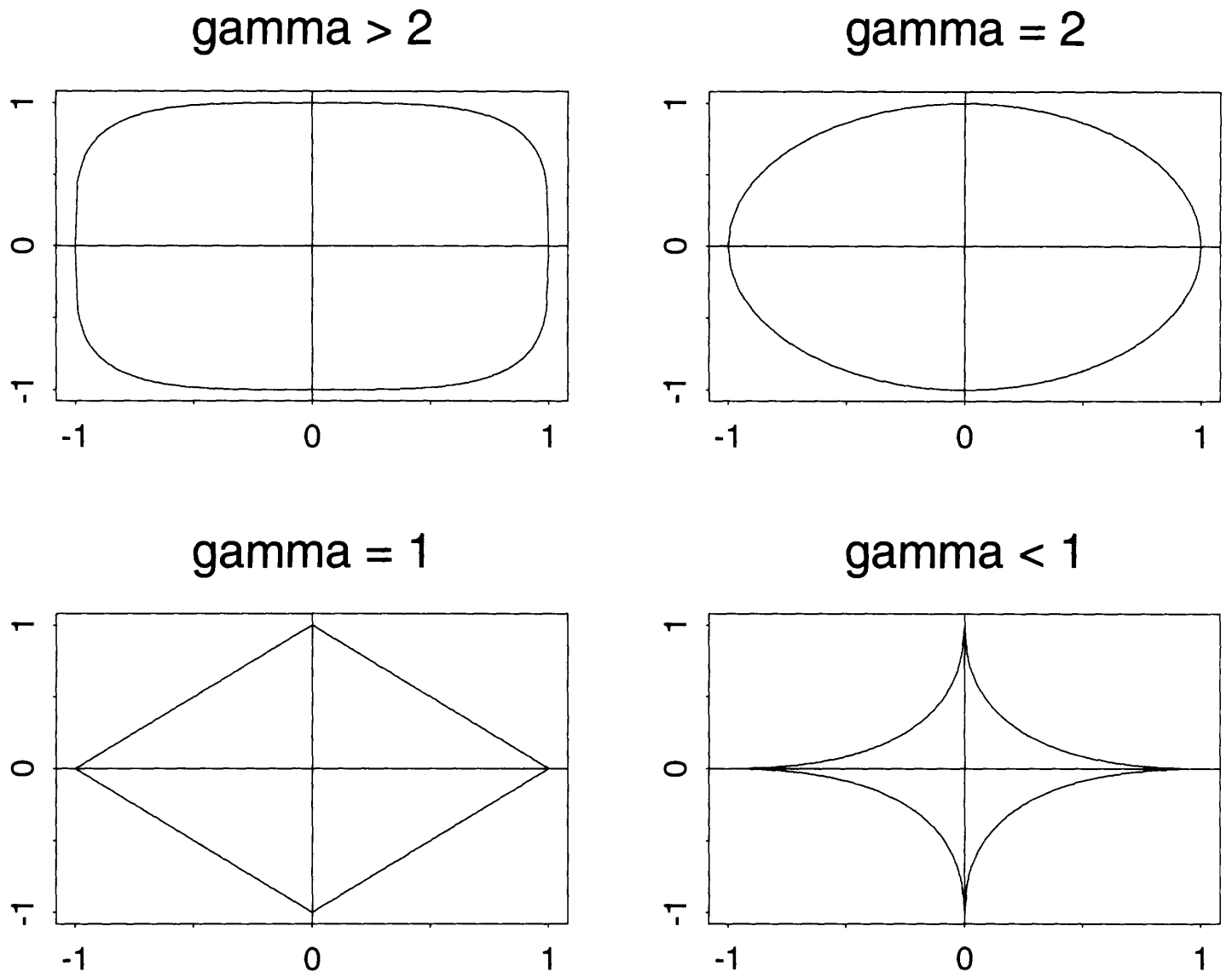


Figure 1. Constrained Areas of Bridge Regressions with $t = 1$.

Nonnegative Garrote

$$\min_c \ \frac{1}{2}\sum_{i=1}^{n}\left(y_i - \sum_{j=1}^{p} c_j \hat{\beta}_j x_{ij}\right)^2 + \lambda \sum_{j=1}^{p} c_j$$

subject to $c_j \geq 0$, and then $\hat{\beta}_j^{ng} = \hat{c}_j \hat{\beta}_j$.

- The resulting estimator is

$$\hat{\beta}_j^{ng} = \left(1 - \frac{\lambda}{2\hat{\beta}_j^2}\right)_+ \hat{\beta}_j.$$

- It is almost unbiased for large $|\hat{\beta}_j|$.
- It shrinks small $|\hat{\beta}_j|$ to zero.

Knight and Fu (AOS; 2000)

Assume $n^{-1}\sum_{i=1}^{n}\mathbf{x}_i\,\mathbf{x}_i^T \to C$ with p.d. C, $n^{-1}\max_i \mathbf{x}_i^T\,\mathbf{x}_i \to 0$, and

$$\Delta(\beta) = (\beta-\beta^*)^T C(\beta-\beta^*) + \lambda_0 \sum_{j=1}^{p} |\beta_j|^r,$$

where β^* is the true regression coefficient.

Theorem
Suppose $\lambda_n/n \to \lambda_0 \geq 0$, *then*

$$\hat{\beta}^{bridge} \xrightarrow{p} \arg\min_{\beta} \Delta(\beta).$$

So if $\lambda_n = o(n)$, *then* $\arg\min_\beta \Delta(\beta) = \beta^*$ *and* $\hat{\beta}^{bridge}$ *is consistent.*

Note: proof is done by Convexity Lemma.

Inconsistent selection of Lasso

Denote $\mathcal{A}^* = \{j : \beta_j^* \neq 0\}$, and $\widehat{\mathcal{A}} = \{j : \hat{\beta}_j^{lasso} \neq 0\}$.

- In general, Lasso is inconsistent in variable selection (Zou, 2006; JASA)

$$\limsup_n P(\widehat{\mathcal{A}} = \mathcal{A}^*) < 1.$$

- Irrepresentable condition (Zhao and Yu, 2006; JMLR): there exists $\mathbf{s} = (s_1, \ldots, s_{p_0})^T$ with $s_j = \pm 1$ such that

$$|\mathbf{C}_{21}\mathbf{C}_{11}^{-1}\,\mathbf{s}\,| \leq 1,$$

  where $\mathbf{C}_{11}$ and $\mathbf{C}_{21}$ are corresponding block matrices in $\mathbf{C}$.

Bach (ICML; 2008)

A more detailed analysis can be done for all possible regularization decays:

- If $\lambda_n \succ n$, then $\hat{\beta}^{lasso} \xrightarrow{p} \mathbf{0}_p$;
- If $\lambda_n \sim n$, then $\hat{\beta}^{lasso} \xrightarrow{p} \arg\min_\beta \Delta(\beta)$;
- If $n^{1/2} \prec \lambda_n \prec n$, then $\hat{\beta}^{lasso} \xrightarrow{p} \beta^*$ and $P(\widehat{\mathcal{A}} = \mathcal{A}^*) \to 1$ under the irrepresentable condition;
- If $\lambda_n \sim n^{1/2}$, then $\hat{\beta}^{lasso} \xrightarrow{p} \beta^*$ and $P(\widehat{\mathcal{A}} \supset \mathcal{A}^*) \to 1$;
- If $\lambda_n \prec n^{1/2}$, then $\hat{\beta}^{lasso} \xrightarrow{p} \beta^*$ and $P(\widehat{\mathcal{A}} = \{1, \ldots, p\}) \to 1$.

Note: here $\succ$, $\prec$ and $\sim$ denote the relative convergence rates.

A second look at the shrinkage methods

One can just focus on the component-wise formulation

$$\min_{\beta} \ \frac{1}{2}(z_j - \beta_j)^2 + J(|\beta_j|)$$

- Hard thresholding: $J(|\beta_j|) = \lambda^2 - (|\beta_j| - \lambda)^2 I(|\beta_j| < \lambda)$, then

$$\hat{\beta}_j = z_j I(|z_j| > \lambda).$$

- Soft thresholding (Lasso): $J(|\beta_j|) = \lambda|\beta_j|$, then

$$\hat{\beta}_j = \mathrm{sign}(z_j)(|z_j| - \lambda)_+.$$

Good penalties? (Fan and Li, 2001; JASA)

The derivative of the componentwise formulation is

$$\text{sign}(\beta_j)(|\beta_j| + J_j'(|\beta_j|)) - z_j.$$

- Unbiasedness: the estimator is nearly unbiased when $|\beta_j|$ is large to avoid model bias.
  - A sufficient condition: $J_j'(|\beta_j|) = 0$ for large $|\beta_j|$
- Sparsity: the estimator sets small estimated coefficient to be exactly zero.
  - A sufficient condition: $\min_{\beta_j} |\beta_j| + J_j'(|\beta_j|) > 0$
- Continuity: the estimator is continuous in z_j to avoid model instability.
  - A nec. and suff. condition: $\arg\min_{\beta_j} |\beta_j| + J_j'(|\beta_j|) = 0$

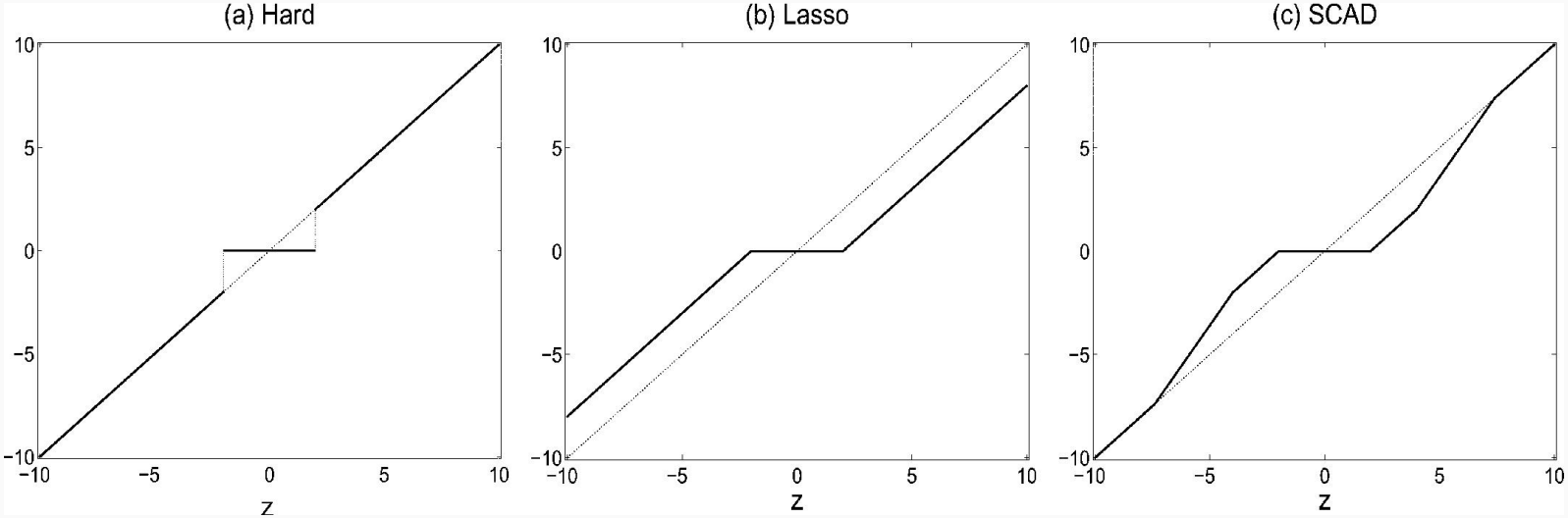
Good penalties?

- Hard thresholding $J(|\beta_j|) = \lambda^2 - (|\beta_j| - \lambda)^2 I(|\beta_j| < \lambda)$ doesn't satisfy the continuity condition
- Soft thresholding $J(|\beta_j|) = \lambda|\beta_j|$ doesn't satisfy the unbiasedness condition

- A good penalty would be truncated lasso penalty

$$J(|\beta_j|) = \lambda\big(\theta - (\theta - |\beta_j|)_+\big),$$

- and some others including the SCAD penalty, the adaptive Lasso penalty, the MCP penalty ...


(a) Hard
(b) Lasso
(c) SCAD
10
5
0
−5
−10
−10
−5
0
5
10
z

SCAD (Fan and Li, 2001; JASA)

$$\min_{\beta} \ \frac{1}{2}\|\mathbf{y} - \mathbf{X}\beta\|^2 + \sum_{j=1}^{p} q_\lambda(|\beta_j|),$$

where

$$q_\lambda(\beta) = \begin{cases} \lambda\beta & \text{if } |\beta| \le \lambda, \\ -\frac{(|\beta|^2 - 2a\lambda|\beta| + \lambda^2)}{2(a-1)} & \text{if } \lambda < |\beta| \le a\lambda, \\ \frac{(a+1)\lambda}{2} & \text{if } |\beta| > a\lambda. \end{cases}$$

- Similar to the truncated lasso penalty, but it is smooth everywhere.

Adaptive Lasso (Zou, 2006; JASA)

$$\min_{\beta} \ \frac{1}{2}\| \mathbf{y} - \mathbf{X}\,\beta\|^2 + \lambda \sum_{j=1}^{p} w_j |\beta_j|,$$

where $w_j > 0$ and adjusts the penalty on each β_j.

- The resulting estimator is

$$\hat{\beta}_j = \mathrm{sign}(\hat{\beta}_j)(|\hat{\beta}_j| - \lambda w_j)_+.$$

- A reasonable choice for w_j is

$$w_j = |\hat{\beta}_j|^{-\xi},$$

  where $\xi > 0$ and $\hat{\beta}$ can be replaced by other root-n consistent estimate for β.

Oracle property

Denote $\beta^1 = (\beta_1, \ldots, \beta_{p_0})^T$ and $\beta^2 = (\beta_{p_0+1}, \ldots, \beta_p)^T$. The truncated Lasso, adaptive Lasso and SCAD all attain the following oracle property.

Theorem
When $\lambda_n/n \to 0$ at certain rate, then with probability tending to 1, $\hat{\beta}^2 = 0$, and

$$\sqrt{n}(I_1(\beta^{*1}) + \Sigma)\left[\hat{\beta}^1 - \beta^{*1} + (I_1(\beta^{*1}) + \Sigma)^{-1} b\right] \to N(0, I_1(\beta^{*1})),$$

where $I_1(\beta^{*1}) = I_1(\beta^{*1}, 0)$, the Fisher information assuming $\beta^{*2} = 0$ is known, and b is first derivative of the penalty term.

Other extensions

- Group lasso: if the p variables are partitioned into J groups, and then it is desirable to include or exclude the whole group

$$\min_{\beta} \ \frac{1}{2}\| \mathbf{y} - \mathbf{X}\,\beta\|^2 + \lambda \sum_{j=1}^{J} \|\beta_j\|_2,$$

  where β_j is a coefficient vector for the j-th group.

- Elastic net: when group structure is not given,

$$\min_{\beta} \ \frac{1}{2}\| \mathbf{y} - \mathbf{X}\,\beta\|^2 + \lambda_1\|\beta\|_1 + \lambda_2\|\beta\|_2^2.$$

- Fused lasso: penalize the difference between the coefficients of connected pairs

Other Methods

Dimension reduction methods

- The methods that we have discussed so far in this module have involved fitting linear regression models, via least squares or a shrunken approach, using the original predictors, $X_1, X_2, \ldots, X_p$.
- We now explore a class of approaches that transform the predictors and then fit a least squares model using the transformed variables. We will refer to these techniques as *dimension reduction* methods.

Assuming additive model...

- Assuming additive model (Shively et al., 1999; Huang and Yang, 2004; Xue, 2009; Huang et al., 2010),

$$y = \sum_{j=1}^{p} f_j^*(x^j) + \epsilon,$$

and then $\mathcal{A}^* = \{j : f_j^* \not\equiv 0\}$

- Lin and Zhang (2006) extends the additive model by allowing higher-order interactions,

$$y = \sum_{j=1}^{p} f_j^*(x^j) + \sum_{j<k} f_{jk}^*(x^j, x^k) + \epsilon$$

and then $\mathcal{A}^* = \{j : f_j^* \not\equiv 0, \text{ or } f_{jk}^* \not\equiv 0 \text{ for some } k\}$

Use of derived input directions I

Principal components regression (PCR): We choose a set of linear combinations of $\mathbf{x}^j$, and then regress $\mathbf{y}$ on these linear combinations.

- The particular combinations used are the sequence of principal components of the inputs. These are uncorrelated and ordered by decreasing variance
- If S is the sample covariance matrix of $\mathbf{x}^1, \ldots, \mathbf{x}^p$, then the eigenvector equations are

$$S\mathbf{q}^t = d_t \mathbf{q}^t,$$

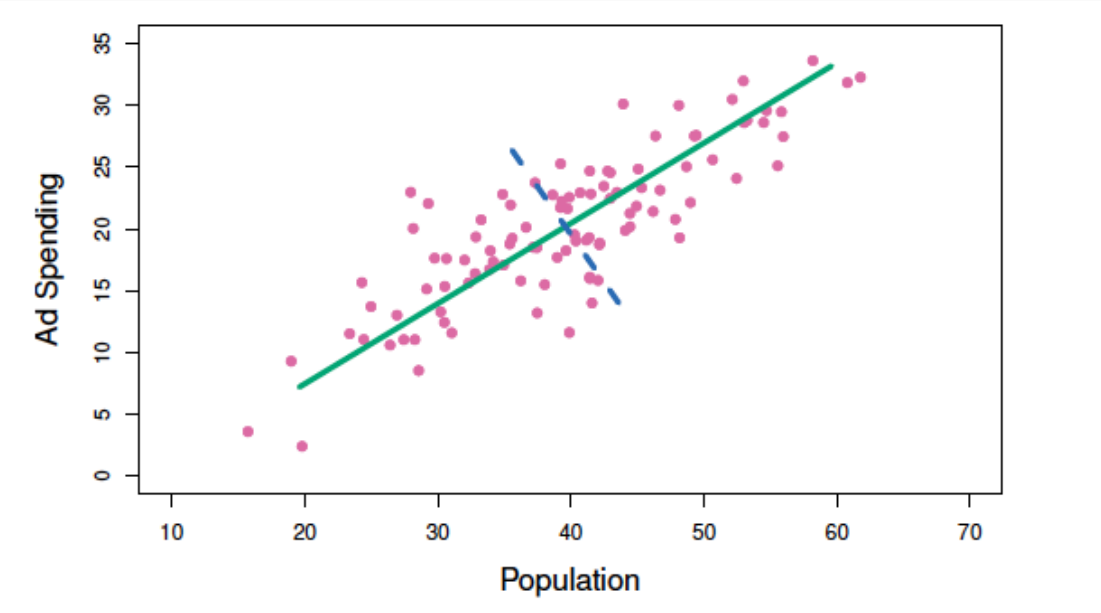
where $d_t \geq 0$ and $\mathbf{q}^t$'s define the principal components of S

- PCR computes the derived input columns $\mathbf{z}^j = \mathbf{X}\,\mathbf{q}^j$ and then regresses $\mathbf{y}$ on $\mathbf{z}^1, \ldots, \mathbf{z}^J$ for some $J < p$.
- Since the $\mathbf{z}^j$'s are orthogonal, this regression is just a sum of univariate regressions,

$$\hat{\mathbf{y}}^{pcr} = \bar{y} + \sum_{j=1}^{J} \hat{\gamma}_j \mathbf{z}^j,$$

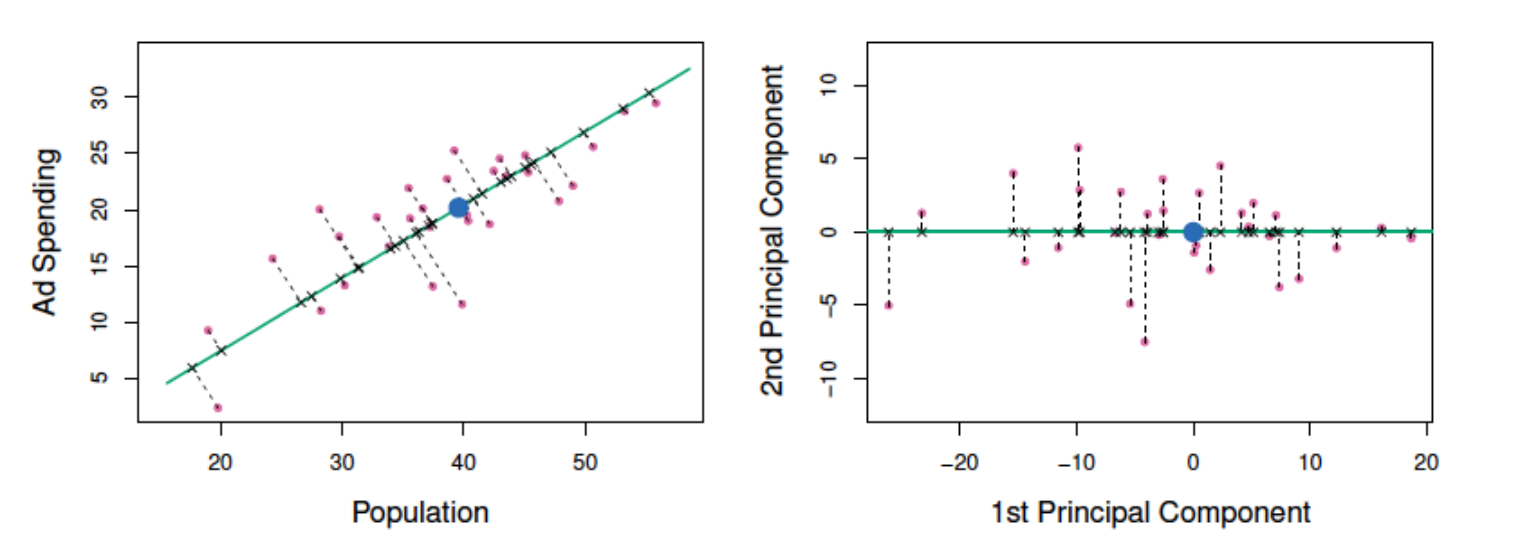
  where $\hat{\gamma}_j$ is the correlation coefficient of $\mathbf{y}$ on $\mathbf{z}^j$.

Pictures of PCA



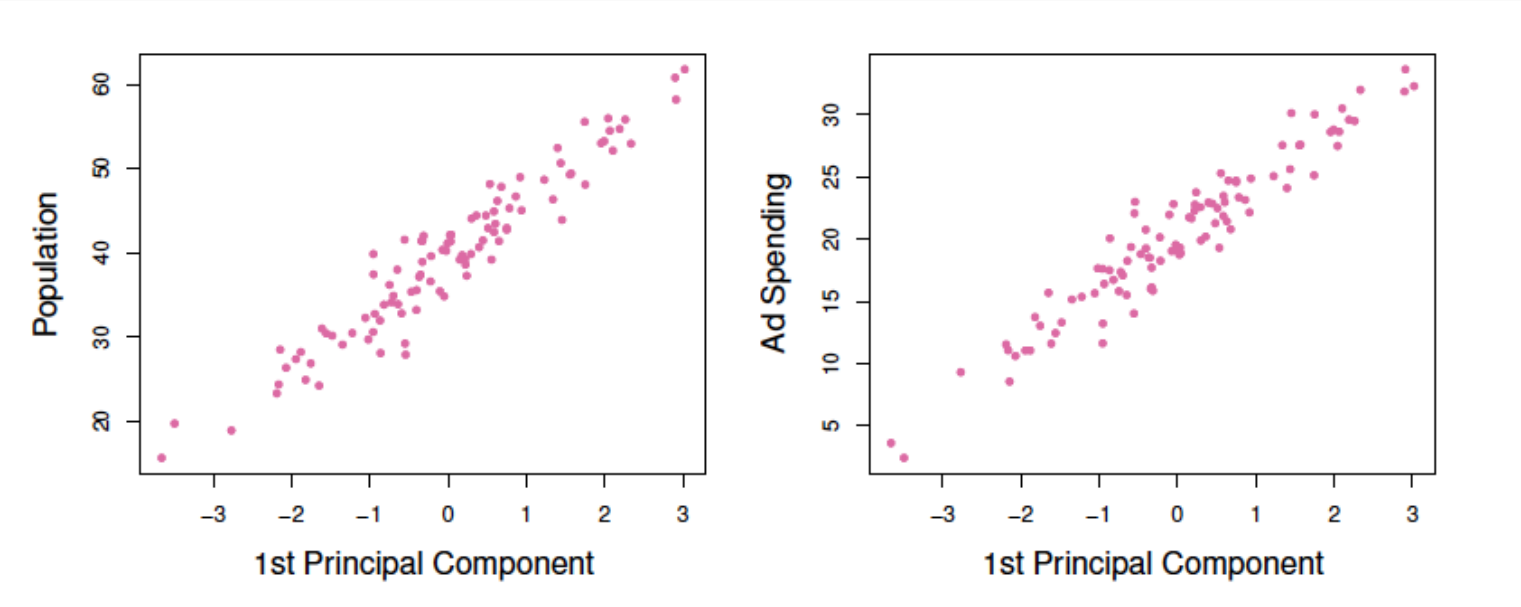
The population size (pop) and ad spending (ad) for 100 different cities are shown as purple circles. The green solid line indicates the first principal component, and the blue dashed line indicates the second principal component.

Pictures of PCA



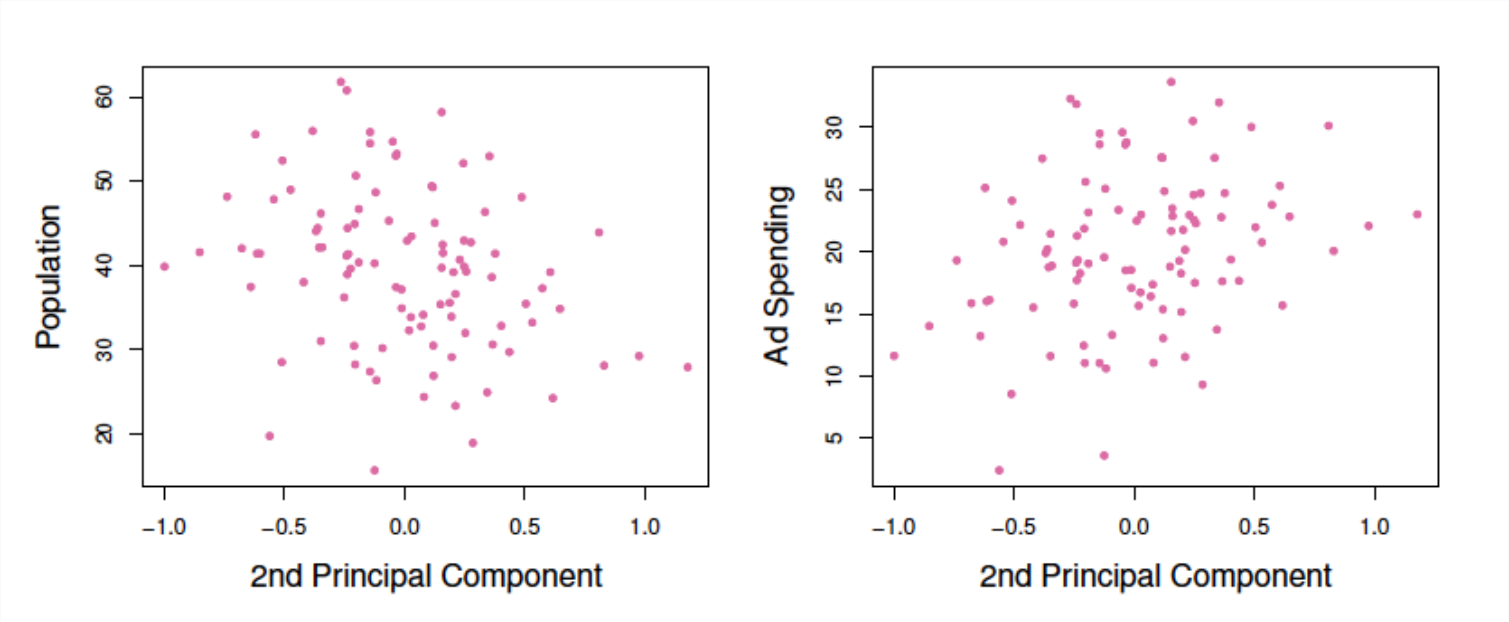
A subset of the advertising data. Left: The first principal component, chosen to minimize the sum of the squared perpendicular distances to each point, is shown in green. These distances are represented using the black dashed line segments. Right: The left-hand panel has been rotated so that the first principal component lies on the x-axis.

Pictures of PCA



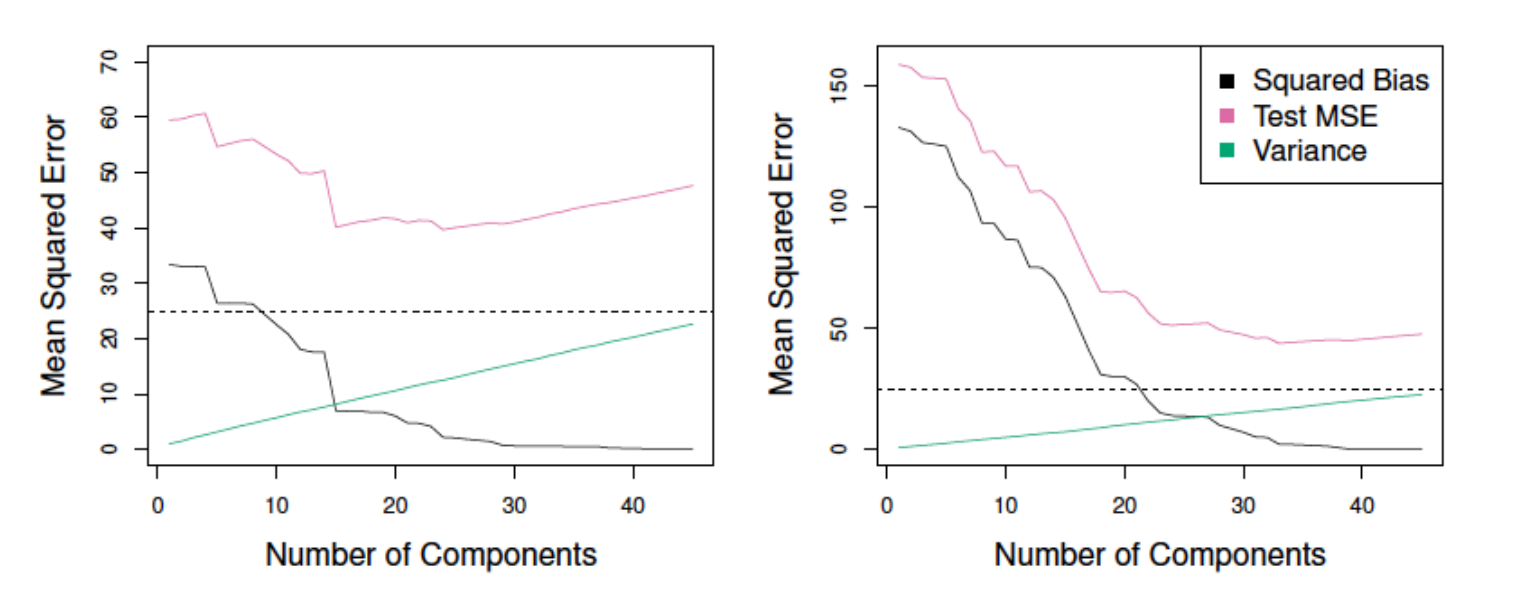
Plots of the first principal component scores z_{i1} versus pop and ad. The relationships are strong.

Pictures of PCA



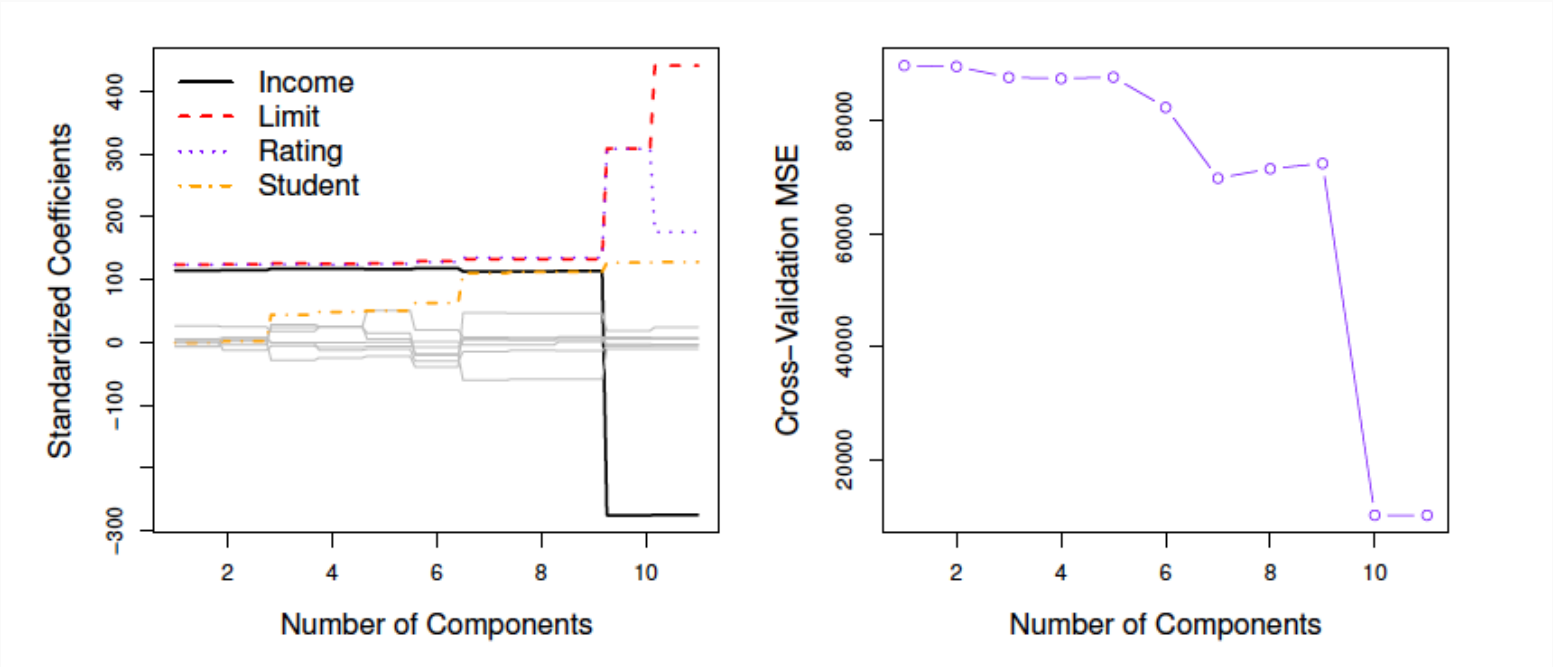
Plots of the second principal component scores z_{i2} versus pop and ad. The relationships are weak.

Application to PCR



PCR was applied to two simulated data sets. The black, green, and purple lines correspond to squared bias, variance, and test mean squared error, respectively. Left: Simulated data I. Right: Simulated data II.

Choosing the number of directions M



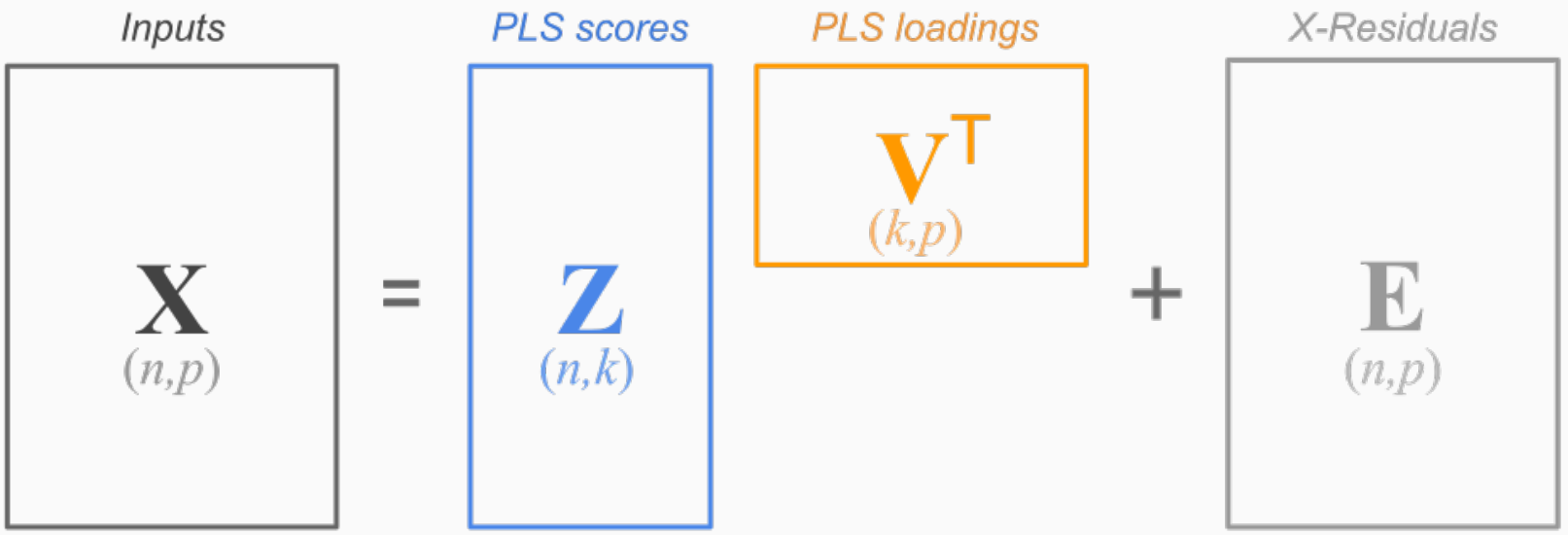
Left: PCR standardized coefficient estimates on the Credit data set for different values of M. Right: The 10-fold cross validation MSE obtained using PCR, as a function of M.

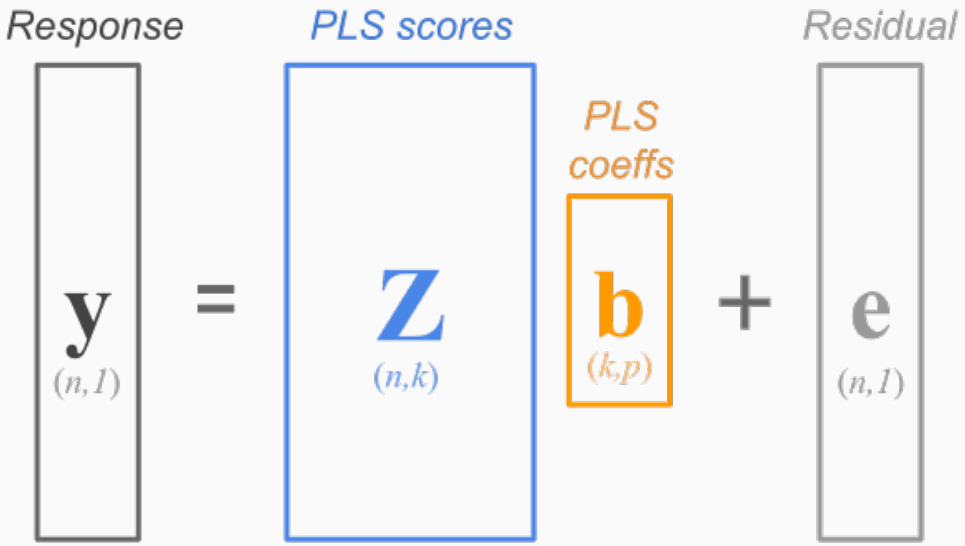
Partial least squares

- PCR identifies linear combinations, or directions, that best represent the predictors $X_1, \ldots, X_p$.
- These directions are identified in an unsupervised way, since the response Y is not used to help determine the principal component directions.
- That is, the response does not supervise the identification of the principal components.
- Consequently, PCR suffers from a potentially serious drawback: there is no guarantee that the directions that best explain the predictors will also be the best directions to use for predicting the response.

- Like PCR, PLS is a dimension reduction method, which first identifies a new set of features $Z_1, \ldots, Z_M$ that are linear combinations of the original features, and then fits a linear model via OLS using these M new features.
- But unlike PCR, PLS identifies these new features in a supervised way — that is, it makes use of the response Y in order to identify new features that not only approximate the old features well, but also that are related to the response.
- Roughly speaking, the PLS approach attempts to find directions that help explain both the response and the predictors.

Some details





By taking a few number of components $Z = [z_1, \ldots, z_k]$, we can have approximations for the inputs and the response. The model for the X-space is:

$$\hat{X} = ZV^T$$

In turn, the prediction model for y is given by:

$$\hat{y} = Zb$$

Then,

$$\hat{y} = Zb^{pls}$$

How does PLS work?

PLS constructs a set of linear combinations of $\mathbf{x}^j$ for regression, but unlike PCR, it uses $\mathbf{y}$ (and $\mathbf{X}$) for this construction.

- Assume $\mathbf{y}$ is centered, and we begin by computing the univariate regression coefficient γ_j of $\mathbf{y}$ on each $\mathbf{x}^j$.
- From this we construct the derived input $\mathbf{z}^1 = \sum \hat{\gamma}_j \mathbf{x}^j$, which is the first PLS direction.
- Then $\mathbf{y}$ is regressed on $\mathbf{z}^1$, giving coefficient $\hat{\beta}_1$; and next we orthogonalize $\mathbf{y}, \mathbf{x}^1, \ldots, \mathbf{x}^p$ with respect to $\mathbf{z}^1$: $\mathbf{r}^1 = \mathbf{y} - \hat{\beta}_1 \mathbf{z}^1$, and $\mathbf{x}^{*j} = \mathbf{x}^j - \theta_j \mathbf{z}^1$.
- We continue this process, until J directions are obtained.

- In this manner, PLS produces a sequence of derived inputs or directions $\mathbf{z}^1, \ldots, \mathbf{z}^J$.
- As with PCR, if we continue on to construct $J = p$ directions, we get back the OLS estimates; use of $J < p$ directions produces a reduced regression.
- Notice that in the construction of each $\mathbf{z}^j$, the inputs are weighted by the strength of their univariate effect on $\mathbf{y}$.

Supervised dimension reduction

- Dimension reduction subspace (DRS): $S_{drs} = S(\eta)$ if

$$y \perp\!\!\!\perp \mathbf{x} \mid \mathbf{x}^T \eta.$$

- Central subspace $S_{y|\mathbf{x}} = \bigcap_\eta S(\eta)$, which is the DRS of the smallest dimension.
- Other DRS's:
  - Mean DRS: $S_{mdrs} = S(\eta)$ if

$$y \perp\!\!\!\perp \mathbb{E}(y|\mathbf{x}) \mid \mathbf{x}^T \eta.$$

  - Discriminant DRS: $S_{ddrs} = S(\eta)$ if

$$\phi_B(y|\mathbf{x}) = \phi_B(y|\mathbf{x}^T \eta),$$

    where ϕ_B is the corresponding Bayes rule.

Inverse regression

Define $S_{\mathbb{E}(\mathbf{x}\,|\,y)} = \text{span}\{\mathbb{E}(\mathbf{x}\,|\,y) - \mathbb{E}(\mathbf{x})\}$, then one can show that

$$S_{\mathbb{E}(\mathbf{x}\,|\,y)} = \text{span}\{\text{var}(\mathbb{E}(\mathbf{x}\,|\,y))\} \subset S_{y|\,\mathbf{x}} = S(\eta),$$

provided that $\mathbb{E}(\mathbf{x}\,|\mathbf{x}^T\eta = u)$ is a linear function in u.

Sliced inverse regression (SIR; Li, 1991):

- slice the dataset according to y_i;
- estimate $\text{var}(\mathbb{E}(\mathbf{x}\,|\,y))$ by the weighted sample covariance matrix based on each slice;
- estimate $S_{y|\,\mathbf{x}}$ by the space spanned by the corresponding eigenvectors of $\widehat{\text{var}}(\mathbb{E}(\mathbf{x}\,|\,y))$.

Minimum average variance estimation (Xia et al., 2002; JRSS-B)

To estimate S_{mdrs}, one can minimize

$$\widehat{\mathbf{B}} = \arg\min_{\mathbf{B}} \sum_{i=1}^{n} L(y_i, f(\mathbf{B}^T \mathbf{x}_i)).$$

subject to $\mathbf{B}^T\mathbf{B} = I_d$, where $f(\mathbf{B}^T\mathbf{x}_i)$ is a low rank approximation of $\mathbb{E}(y|\,\mathbf{x}_i)$.

In specific, MAVE proposes to use local linear fitting as f and solves

$$\underset{\alpha,\beta,\mathbf{B}}{\operatorname{argmin}} \sum_{i=1}^{n}\sum_{j=1}^{n} w_{ij}\Big(y_i - (\alpha_j + \beta_j \mathbf{B}^T(\mathbf{x}_i - \mathbf{x}_j))\Big)^2.$$

References

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The Elements of Statistical Learning: Data Mining, Inference, and Prediction, Trevor Hastie, Robert Tibshirani, Jerome Friedman, Springer, 2008